Training-Free Cultural Alignment of Large Language Models via Persona Disagreement

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Abstract

LLMs are increasingly deployed in decisions that require culture-dependent moral judgements, yet they answer as if the whole world thinks with a Western mindset. The Moral Machine experiment [Awad et al., 2018] showed this is wrong at scale: 40 million judgments across 233 countries reveal that moral preferences are systematically structured by culture, and a model that ignores this variation does not merely underperform, but also imposes one society's intuitions on all others. Existing fixes do not scale to global deployment, as fine-tuning needs per-country preference data and GPU budgets, reward-guided decoding needs per-country reward models, and activation steering needs access to model internals that black-box APIs do not expose. In this work, we focus on this realistic inference time regime, with no weight updates, no training data, and no internal access. The key observation is that within-country demographic *disagreement*, not consensus, is the steering signal. When culturally grounded personas agree, the base model is already calibrated. But when they disagree, the spread tells us what to fix and how. We propose DISCA, which instantiates each country as a panel of four World-Values-Survey-grounded persona agents [Haerpfer et al., 2020]. It converts their disagreement into a bounded, loss-averse correction, whose magnitude is set by the panel's variance, and shrinks the correction toward zero when the estimate is unreliable. Across 20 countries and 7 open-weight backbones from five model families, DISCA reduces cultural misalignment on MultiTP [Jin et al., 2025] by 10–24% on the six $\geq$3.8B backbones, 3.4% on the smallest 2B model on binary moral dilemmas, and 2–7% on open-ended scenarios. Furthermore, a 14B backbone with DISCA reaches lower absolute misalignment than a vanilla 70B model.

1 Introduction

Large language models increasingly mediate decisions that turn on moral judgement: content moderation policies, clinical prioritisation guidelines, autonomous-vehicle behaviour specifications, and high-stakes recommendation pipelines. In each of these settings, the model's implicit moral preferences become the system's moral preferences. Yet a growing body of evidence shows that these implicit preferences are not culturally neutral. The Moral Machine experiment collected 40 million judgments across 233 countries and showed that moral preferences are systematically structured by culture [Awad et al., 2018]; large language model (LLM) outputs correlate most strongly with respondents from Western, educated, industrialised, rich, and democratic populations [Henrich et al., 2010, Santurkar et al., 2023]; and post-training alignment introduces further unintended cultural shifts [Ryan et al., 2024, Zewail et al., 2026]. A model that ignores this variation does not merely underperform on a benchmark, but also imposes one society's intuitions on every user it serves.

The cultural gap is now well-documented at the model level. The MultiTP benchmark [Jin et al., 2025] extends the Moral Machine protocol to 107 languages with country-specific Average Marginal Component Effects (AMCEs), and reports consistently weak country-level alignment across all 19

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evaluated LLMs. The asymmetries are substantial, e.g., Japan ranks first among countries on preference for sparing pedestrians, while China ranks 116th, and current models do not capture either extreme. Concurrent work shows that sociodemographic persona prompts shift LLM moral decisions two to four times more than they shift human decisions [Kim et al., 2025], indicating that the models are not insensitive to cultural framing, but that the framing they receive at deployment time does not match the cultural distribution they are asked to serve.

Closing this gap is harder than it looks. Three families of methods have been proposed, and each demands resources that do not scale to the long tail of countries served by a single deployed model. Fine-tuning and culture-aware adapters require per-country preference datasets and GPU budgets for every target population [Zhang et al., 2026, Yao et al., 2025]. Reward-guided decoding requires a separate trained reward model per country [Khanov et al., 2024, Mudgal et al., 2024], and activation steering requires write-access to model internals [Arditi et al., 2025, Turner et al., 2024] that black-box APIs do not expose. Any deployable solution must therefore work at inference time, must not assume per-country reward models or labelled preference data, and must not assume write access to model internals beyond what an API exposes. We refer to this as the *black-box, public-data-only regime*: the method we propose requires only API-level access to decision-token log-probabilities (exposed by every open-weight backbone we evaluate and by major commercial APIs that return `logprobs`), and we argue this is the *only regime where cultural alignment can scale from a handful of curated countries to the hundreds a globally-deployed model actually serves*.

We observe that when certain culturally grounded persona prompts representing the same country face the same moral dilemma, the dispersion of their decoding-time logit gaps is itself informative. If they agree, the base model is already calibrated for that country and no correction is needed. But if they disagree, the spread of their gaps encodes both the direction and the magnitude of the needed correction. Our method, DISCA (Disagreement-Informed Steering for Cultural Alignment), instantiates each country as a panel of four persona agents grounded in World Values Survey microdata [Haerpfer et al., 2020], evaluates them in a single batched forward pass, and converts the panel's variance into a bounded, loss-averse logit correction whose magnitude is automatically attenuated when the underlying estimate is unreliable. We show that this attenuation is principled: the underlying oracle correction is the MSE-optimal scalar shrinkage of the consensus, whose weight depends only on the within-panel variance (Proposition 1), and we approximate it with an empirical variance-aware shrinkage heuristic.

Across 20 countries spanning four continents and seven open-weight backbones (2B–70B parameters) from five model families, DISCA reduces cultural misalignment on MultiTP by 10–24% on the six $\geq$3.8B backbones (and 3.4% on the smallest 2B model) on binary moral dilemmas, and 2–7% on open-ended scenarios. The strongest absolute alignment is achieved by Phi-4 (14B), which surpasses a vanilla Llama-3.3-70B backbone despite using one fifth of the parameters; this suggests that, in the regime we study, calibration competes with scale rather than merely complementing it. An evaluation on the BLEnD factual cultural QA benchmark [Myung et al., 2024] reveals a clean scope boundary: DISCA steers values, where the decision reduces to a scalar logit gap, but does not transfer to factual recall, where the decision is a single token in a large vocabulary. Failure modes are architecturally rather than culturally determined and are diagnosable from the base model's vanilla logit conditioning alone. Primary metrics (MIS and diagnostics) are defined in §4. Our contributions are summarised as follows:

- An inference-time cultural alignment method that requires no weight updates, no per-country reward models, and no white-box access.
- A formal characterisation of disagreement-driven shrinkage. DISCA is the first inference-time alignment method to use independent-run disagreement as a reliability signal.
- An empirical evaluation across 20 countries, four continents, seven open-weight backbones, and three evaluation formats (binary moral dilemmas, open-ended scenarios, factual QA).

2 Related Work

Benchmarks and cultural evaluation. The Moral Machine experiment [Awad et al., 2018] established that moral preferences are culturally structured, and MultiTP [Jin et al., 2025] extends it to 107 languages with country-specific AMCEs; LLM alignment to these AMCEs varies widely across model families [Takemoto, 2024, Ahmad and Takemoto, 2025] and correlates most with WEIRD respondents [Santurkar et al., 2023, Henrich et al., 2010], with RLHF adding unintended cultural

shifts [Ryan et al., 2024, Zewail et al., 2026]. Kim et al. [2025] further show that sociodemographic personas shift LLM moral decisions 2–4$\times$ more than they shift humans (a "partisan sorting" phenomenon absent in human respondents). This last finding directly motivates our design: rather than using personas to steer toward a single viewpoint, DISCA uses them to *measure* within-country disagreement and converts that into a bounded correction.

Inference-time alignment. The pluralistic alignment vision [Sorensen et al., 2024] argues that inference-time methods are a natural vehicle for cultural adaptation, but existing approaches each sacrifice one of the three constraints we target: activation steering [Arditi et al., 2025, Turner et al., 2024] requires white-box access; reward-guided decoding [Khanov et al., 2024, Mudgal et al., 2024] needs per-country reward models; culture-aware adapters [Zhang et al., 2026, Yao et al., 2025] need per-culture training. Training-time prospect-theoretic alignment [Ethayarajh et al., 2024] uses the same PT value shape we adopt, but at training time; DISCA applies it at *test time*, per scenario, aggregating heterogeneous personas rather than shaping a trained policy.

Distinction from superficially similar paradigms. DISCA may resemble ensemble methods architecturally, but not algorithmically. In self-consistency [Wang et al., 2023] and LLM debate [Du et al., 2024, Liang et al., 2024], diversity is noise to be eliminated via majority vote; standard ensembles average outputs to reduce *prediction* variance. In DISCA, diversity *is* the signal: the *spread* of persona logit gaps, not the mode, sets the correction magnitude. We show (Proposition 1) that the MSE-optimal scalar shrinkage of the consensus correction is a closed-form function of within-panel variance, making within-panel variance a *sufficient statistic for correction reliability*. Standard calibration (temperature scaling, MC-Dropout [Gal and Ghahramani, 2016, Kwon et al., 2026]) adjusts confidence isotropically; DISCA applies anisotropic, loss-averse corrections. DISCA is the first training-free method targeting country-level moral calibration; the method is grounded in control-as-inference, Prospect Theory, and importance-sampling variance control.

3 Disagreement-Informed Steering for Cultural Alignment

DISCA aligns the model by creating an adjustment derived from the agents' consensus, and the agreement among the agents controls how much of it actually applies. The magnitude of the adjustment is not fixed: when the agents agree on what the preference, we apply the adjustment in full; when they disagree, we apply only a fraction of it, scaled by how much the agents disagree. This inverts the standard combination structure used by self-consistency [Wang et al., 2023], multi-agent debate [Du et al., 2024, Liang et al., 2024], and ensembles, where N candidate outputs are aggregated by majority or averaging and disagreement among them is treated as noise to be suppressed. DISCA instead treats disagreement as the quantity that controls how much adjustment to apply.

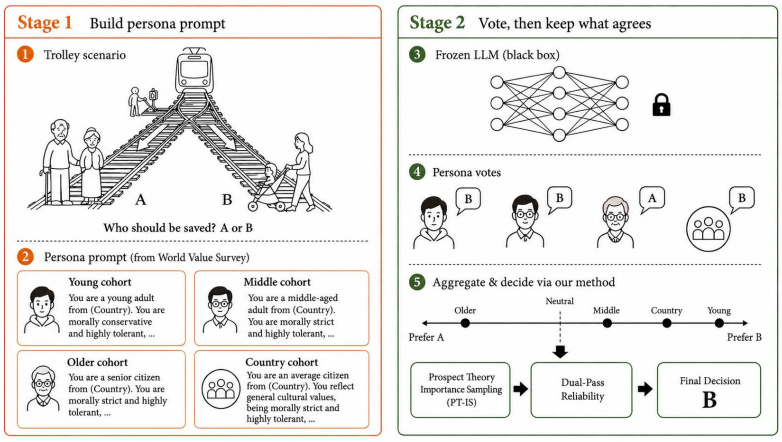


Figure 1: **DISCA overview.** Stage 1 builds WVS-grounded persona prompts for a trolley scenario in country c; Stage 2 runs a frozen large language model (LLM) on the base prompt and each persona, aggregates persona-level signals in logit space, and applies *Prospect-Theory importance sampling* (PT–IS) together with a dual-pass reliability gate to obtain the final sparing probability. Pseudocode and the six MultiTP attribute–temperature pairs provided in App. A1.

3.1 Setup

We consider a frozen language model f_θ on a forced-choice task: each input x asks the model to choose between two options via decision tokens A and B. Let $z(x) = [z_a, z_b]$ denote the model's logits at the decision-token position, and let $\delta(x) = z_b - z_a$ be the resulting decision gap. For target country c, we assume access only to f_θ as black-box and a public characterisation of country c's population (in our case, World Values Survey microdata [Haerpfer et al., 2020]); we do not assume per-country preference labels, reward models, or write access to θ. Our goal is to compute, from f_θ alone, a per-scenario logit adjustment $\delta^\star(x, c)$ such that the adjusted gap $\delta(x) + \delta^\star(x, c)$ produces decisions better aligned with country c's preferences. Alignment is measured against a country-level reference vector; the benchmark and metric are described in §4. The remainder of Section 3 focuses on how $\delta^\star$ should depend on the disagreement, what formal property this dependence has, and how to estimate the correction.

3.2 The disagreement principle

Suppose we evaluate the same scenario under N culturally grounded prompts representing the same target country, each producing a logit gap δ_i. Each δ_i is a noisy estimate of what country c's population-level preference would imply for this scenario. If the gaps cluster tightly, the N estimates converge on the same answer and the consensus is a reliable signal. If they are dispersed, the consensus is averaging over disagreeing estimates and is unreliable as a steering target. The principle is that within-agent dispersion, not consensus magnitude, controls how much of the consensus adjustment we should actually apply.

Let $\delta_1, \ldots, \delta_N$ be the logit gaps produced by N agents on a single scenario, and let δ_h denote the unobserved logit gap consistent with country c's population-level preference on that scenario. We model the agents as $\delta_i = \delta_h + \eta_i,\ \mathbb{E}[\eta_i] = 0,\ \mathrm{Var}(\eta_i) = \tau^2$, with the η_i assumed independent across agents and identically distributed with common variance τ^2. This is a standard hierarchical setup: δ_h is a fixed-but-unknown population parameter, and each agent is a noisy view of it. The i.i.d. structure is a modelling simplification, since all agents share base model f_θ and the scenario. Define the consensus $\bar{\delta} = \frac{1}{N}\sum_i \delta_i$, the within-agent variance $D^2 = \frac{1}{N-1}\sum_i (\delta_i - \bar{\delta})^2$, the consensus adjustment $\Delta = \bar{\delta} - \delta_{\text{base}}$, and the oracle adjustment $\Delta_h = \delta_h - \delta_{\text{base}}$, where δ_{base} is the logit gap produced by the unconditioned base prompt (without persona or country-related prompts) on the same scenario. We will apply a scaled version of Δ as our adjustment: $\hat{\Delta}(\gamma) = \gamma \cdot \Delta$ for some $\gamma \in [0, 1]$. We choose γ by minimising the mean squared error against the unobserved oracle: $\mathrm{MSE}(\gamma) = \mathbb{E}\big[\big(\hat{\Delta}(\gamma) - \Delta_h\big)^2\big]$.

Proposition 1 (Variance-aware shrinkage - Proof in App. A2)**.** *Under the agent model above:*

(i) *D^2 is unbiased for τ^2, and the variance of the consensus adjustment is $\mathrm{Var}(\Delta) = \tau^2/N$.*

(ii) *Among shrinkage estimators of the form $\hat{\Delta}(\gamma) = \gamma \cdot \Delta$ with $\gamma \in [0, 1]$, the MSE-minimising one has $\gamma^\star = \frac{\Delta_h^2}{\Delta_h^2 + \tau^2/N}$, which lies in $[0, 1]$ and is monotone-decreasing in τ^2.*

3.3 The DISCA estimator

Proposition 1 tells us what shape the shrinkage should have but not how to compute it: $\gamma^\star$ depends on the unobservable Δ_h. In particular, the proposition's two parts work together to motivate the method. Part (ii) gives the optimal $\gamma^\star$ in closed form, but it is not directly computable: $\gamma^\star$ depends on τ^2 and on Δ_h, neither of which we observe. Part (i) gives us a way to estimate τ^2 from data we already have: the within-agent variance D^2 has expected value equal to τ^2. As Δ_h has no analogous estimator from the agents' outputs, we cannot compute $\gamma^\star$ exactly. Instead, we construct an estimator $\delta^\star$ that is computable from the agents' outputs alone, matches the shape of $\gamma^\star$, bounded in $(0, 1)$, & monotone-decreasing in the variance estimate.

We use $N{=}4$ agents per target country: three age cohorts (young, middle-aged, older) plus a country-wide aggregate, each instantiated as a native-language system prompt built from cohort- and country-level aggregates of the WVS cultural profile of Greco et al. [2026] (full prompts in App. A12). The choice $N{=}4$ is the smallest panel that simultaneously covers the three demographic axes documented to carry most within-country variance [Inglehart and Welzel, 2005] and includes a country-wide aggregate that anchors the ensemble against cohort-specific sampling noise; the marginal precision gain from each additional persona decays as $\Theta(N^{-2})$ (Corollary 1,

App. A2), placing $N{=}4$ on the flat side of the diminishing-returns curve. The empirical sweep over $N \in \{2, 3, 4, 5, 6\}$ on a three-country panel (Table 22) confirms this: $N{=}4$ minimises macro MIS, $N{=}2$ leaves the consensus under-determined, and $N{\geq}5$ duplicates demographic coverage without adding new variance. To remove the order bias of forced-choice decoding, we evaluate every scenario under both (A, B) and (B, A) orderings and take the order-symmetrised gap $\delta_i = \frac{1}{2}(\delta_i^{(AB)} - \delta_i^{(BA)})$ as the agent's output.

3.3.1 Loss-averse importance sampling

The plain consensus adjustment $\Delta = \bar{\delta} - \delta_{\text{base}}$ moves the base model toward the average direction because $\bar{\delta}$ is the average consensus. The problem is that an average direction can hide a cohort that strongly disagrees: three cohorts mildly in favour and one cohort strongly against will average out to "mildly in favour," and the consensus will move the model in a direction that one cohort actively rejects. We want a steering rule that avoids this. One that prefers a direction nobody strongly dislikes over a direction with a higher average but a strong dissenter.

We frame this as a cooperative-bargaining problem among the cohorts [Nash, 1950, Kalai and Smorodinsky, 1975]. Each cohort is a player with its own utility over correction directions, each is loss-averse, which means a small misalignment with one's preferred direction is felt more than a same-sized improvement, and we want to pick a direction that all players can collectively accept. This framing organises three design choices: where to look for candidates, how each player scores a candidate, and how all scores combine into a group verdict.

Rather than committing to Δ, we draw K candidate perturbations $\epsilon_k \sim \mathcal{N}(0, \sigma^2)$ in a neighbourhood of the consensus, defining the candidate state $\tilde{\delta}_k = \bar{\delta} + \epsilon_k$. Each candidate is scored by how much closer to cohort i's preferred direction it lands, in absolute ℓ_1 distance, against two reference points: the cohort gap δ_i and the panel consensus $\bar{\delta}$,

$$g_{i,k} = |\delta_{\text{base}} - \delta_i| - |\tilde{\delta}_k - \delta_i|, \qquad g_{\text{cons},k} = |\delta_{\text{base}} - \bar{\delta}| - |\tilde{\delta}_k - \bar{\delta}|, \tag{1}$$

so positive g means the candidate moved closer to the relevant target than the base prompt did, negative means it moved the model away.

We need a scoring rule $v(\cdot)$ that is concave on the gain side so additional gain beyond what's already aligned is worth less and steeper on the loss side so any cohort loss matters more than the same-size gain elsewhere. The canonical such rule in the decision-theory literature is the Kahneman–Tversky value function [Kahneman and Tversky, 1979, Tversky and Kahneman, 1992]:

$$v(z) \;=\; \begin{cases} z^{\alpha} & z \geq 0, \\ -\kappa\,(-z)^{\alpha} & z < 0, \end{cases} \tag{2}$$

with curvature $\alpha \in (0, 1]$ and loss-aversion $\kappa \geq 1$. The asymmetric kernel v acts as an aggregator that penalises misalignment more than rewarding incremental alignment.

Cooperative bargaining sits between two extremes: a purely *utilitarian* verdict averages the players' utilities and ignores how the gains are distributed, while a purely *collective* verdict asks how the deal serves the group as a whole. We use a blend of the two:

$$U_{\text{total}}(\epsilon_k) \;=\; (1 - \lambda_{\text{coop}})\,\frac{1}{N}\sum_{i=1}^{N} v\Big(\frac{g_{i,k}}{\sigma}\Big) \;+\; \lambda_{\text{coop}}\, v\Big(\frac{g_{\text{cons},k}}{\sigma}\Big), \tag{3}$$

where $g_{\text{cons},k}$ measures the deal's gain against the consensus direction $\bar{\delta}$ rather than against any single cohort. The first term is the utilitarian verdict over loss-averse players; the second is the collective verdict scored by the same loss-averse rule. The coefficient $\lambda_{\text{coop}} \in [0, 1]$ controls the bargaining position, with $\lambda_{\text{coop}} = 0$ purely utilitarian and $\lambda_{\text{coop}} = 1$ purely collective. We use $\lambda_{\text{coop}} = 0.7$, on the collective side; a sensitivity sweep over $\lambda_{\text{coop}} \in [0, 1]$ is in App. A11. The division by σ inside v scores each gain in units of the perturbation scale, so the search radius σ and the value-function shape (α, κ) can be set independently.

The natural choice, returning the highest-utility candidate, is unsafe here. The utility $U_{\text{total}}(\epsilon_k)$ is itself a noisy quantity. It depends on the random draw ϵ_k and on the agent gains $g_{i,k}$, both of which carry stochastic error. Picking the argmax over K noisy scores selects the candidate that drew favourable noise on this sample, not the candidate with the highest true utility. We therefore

aggregate all candidates with weights proportional to their utility:

$$\delta_{PT-IS} \;=\; \frac{\sum_{k=1}^{K} w_k\,\epsilon_k}{\sum_{k=1}^{K} w_k}, \qquad w_k \propto \exp\!\big(U_{\text{total}}(\epsilon_k)/\eta\big), \tag{4}$$

where $\eta > 0$ is an IS softmax temperature (η=0.5 default; Table 19). This form is not ad hoc: it is the path-integral solution to the stochastic optimal control problem of selecting a correction direction under a noisy utility function [Williams et al., 2018, Levine, 2018]. The high-utility candidate dominates the average, but near-winners contribute as well, and stochastic error in any one candidate's utility is suppressed by the contributions of the others. We refer to the full procedure as *loss-averse importance sampling*, or PT-IS for short.

3.3.2 Dual-pass reliability gate

PT-IS gives us a single correction $\delta_{\text{PT-IS}}$ for each scenario. The natural next step is to apply it: add $\delta_{\text{PT-IS}}$ to the model's base logits and read off the steered decision. The problem is that $\delta_{\text{PT-IS}}$ is a single number combining cohort opinions. When all cohorts agree, that number reflects a unanimously backed direction. When the cohorts disagree, the same number can come out small even though no cohort actually wants the small push it represents.

Concretely, imagine two cohorts prefer direction $+0.5$ and two prefer -0.5. PT-IS with loss-aversion will not pick either side; it returns a compromise, say $\delta_{\text{PT-IS}} = +0.05$, close to zero because pushing in either direction would hurt half the cohorts. Applying $+0.05$ to the model's logits nudges the model in a direction no cohort actually asked for. This is the failure mode Proposition 1 warned us about: $\gamma^\star$ contracts toward zero when cohort disagreement is high, because the appropriate response to disagreement is to apply less correction, not a more carefully averaged correction. We need a way to identify contested scenarios at run time and contract $\delta_{\text{PT-IS}}$ accordingly.

The obvious approach is to increase K. With infinitely many candidates, $\delta_{\text{PT-IS}}$ converges to a deterministic limit and run-to-run variation vanishes. But this does not solve the problem: the $+0.05$ compromise becomes a more precise compromise. More samples sharpen the average; they do not change whether the average deserves to be applied at full strength.

We run PT-IS twice on the same scenario, on disjoint sub-budgets of size $K/2$, producing independent corrections $\delta^{(1)}_{\text{PT-IS}}$ and $\delta^{(2)}_{\text{PT-IS}}$. When the cohorts agree, both runs find the same compromise and the two corrections are close. When the cohorts disagree, the runs are pulled by competing cohort camps and land on different compromises. The squared inter-run gap

$$V_r \;=\; (\delta^{(1)}_{\text{PT-IS}} - \delta^{(2)}_{\text{PT-IS}})^2$$

is the diagnostic: zero when the runs agree, growing as they diverge. V_r is the half-sample variance estimator: a single sample is split in two, and variance is estimated from the squared inter-half gap. The technique originates in Mahalanobis [**?**] and was generalised as balanced repeated replication [**??**]; it is canonical in settings that need a qualitative variance signal on a constrained compute budget. Two runs also keep total compute matched to a single full PT-IS run.

We multiplicatively shrink the correction by

$$r \;=\; \exp\!\big(-V_r/s\big) \;\in\; (0, 1], \tag{5}$$

with bandwidth $s > 0$ controlling the decay rate. The exponential is the simplest function satisfying the three shape properties Proposition 1 requires: bounded in $(0, 1]$, monotone-decreasing, and smooth. The final DISCA correction is the gated average of the two runs:

$$\delta^\star \;=\; r \cdot \frac{\delta^{(1)}_{\text{PT-IS}} + \delta^{(2)}_{\text{PT-IS}}}{2}. \tag{6}$$

$\delta^\star$ matches the qualitative shape of $\gamma^\star \cdot \Delta$: bounded by the average run magnitude, monotone-decreasing in inter-run disagreement (mirroring $\gamma^\star$ in τ^2), and smooth in all inputs. The empirical contribution of the gate is bundled inside the full-method ablation in Table 4.

3.4 Extension to Open-Ended Ethical Scenarios

To extend DISCA to open-ended scenarios, the model generates free-form responses under both the original prompt and four persona-specific prompts. A separate large language model (LLM) judge then parses these outputs to return a {choice, confidence} pair. The continuous decision signal is represented by a pseudo-logit gap (the A/B logit distance) extracted by that judge. The dual-pass Prospect-Theory importance sampling (PT–IS) mechanism is applied similarly to the binary case.

4 Experimental Results

We test DISCA on MultiTP [Jin et al., 2025], the multilingual extension of the Moral Machine trolley problem [Awad et al., 2018]. Each scenario shows the model two groups of people that an autonomous vehicle could spare, and the model has to choose one of them by emitting a single decision token, A or B. Every scenario isolates one of six moral attributes (species, gender, age, fitness, social value, and the number of lives saved) while matching the two groups on everything else. An age scenario, for example, pits a young group against an older one and holds the rest of the description constant; the rate at which the model spares the younger group, averaged across many such scenarios, is then read off as its preference on age. Following the Moral Machine convention we always assign decision token B to the side the AMCE measures toward; this is a measurement convention from conjoint analysis that fixes the sign of the preference, not a normative claim about which side ought to be saved. The frozen model emits logits $z(x) = [z_a, z_b]$ at the decision position, the decision gap is $\delta(x) = z_b - z_a$, and the sparing probability under decoding temperature T_{dec} is $p_{\text{spare}}(x) = \sigma(\delta(x)/T_{\text{dec}})$. Averaging the sparing probability across the scenarios for one attribute and one country gives that country's Average Marginal Component Effect (AMCE):

$$\hat{m}_d^{(c)} = \frac{1}{|\mathcal{S}_d^{(c)}|} \sum_{x \in \mathcal{S}_d^{(c)}} p_{\text{spare}}(x). \tag{7}$$

Stacking the six attributes gives a six-dimensional vector $\hat{m}^{(c)}$ that we compare to the human AMCE vector $h^{(c)}$ aggregated from real Moral Machine judgments [Awad et al., 2018, Jin et al., 2025]. Our headline metric is the *misalignment score* $\text{MIS}(c) = \|\hat{m}^{(c)} - h^{(c)}\|_2$, which is just the straight-line distance between the model's preference vector and the country's human one; lower is better. We report it macro-averaged over a 20-country panel that spans four continents. The country list, the preprocessing pipeline (deduplication, quality filter, per-attribute caps, oversampling, and deterministic shuffling), the per-country slice sizes, and a cap-disabled sensitivity rerun all live in App. A14. Every scenario is presented in the country's native language; only the decision tokens A and B stay in English so that token-ID extraction is consistent across model families. We also report Pearson r for shape agreement and Jensen–Shannon divergence (JSD) as secondary diagnostics.

We tested 28 model–method combinations across 12 architectures from 270M to 70B parameters, with the full landscape in App. A7, and report seven open-weight backbones with the most robust gains: Llama-3.3-70B, Magistral-24B, Phi-4 (14B), Qwen3-VL-8B, Qwen2.5-7B, Phi-3.5-mini (3.8B), and Gemma-4-E2B (2B). The few models that degrade share a single pattern: their vanilla MIS is already low enough that there is no headroom left for any correction (we return to this in §4.2). The released hyperparameter configuration was validated on a small three-backbone, five-country prototyping panel before being frozen for the headline sweep; how we elicit the A and B tokens consistently across model families is described in App. A13.

We compare DISCA against two tiers of baselines on the same 20-country Phi-4 grid (Table 1; full implementations in App. A10). The first tier is training-free and never sees the human AMCE: vanilla decoding, a WVS Profile Prompt that summarises the country's WVS values, a PRISM-style cultural framing prompt [Kirk et al., 2024], a fixed logit offset, activation steering [Arditi et al., 2025, Zou et al., 2023], and MC-Dropout [Gal and Ghahramani, 2016, Kwon et al., 2026]. We exclude reward-guided decoding [Khanov et al., 2024, Mudgal et al., 2024] because it assumes per-country reward models, which simply do not exist for the long tail of countries we care about. The second tier is the opposite: oracle baselines that fit one per-country scalar (temperature or additive margin) directly on the country's full human AMCE [Guo et al., 2017], serving as upper bounds on what any 1-d-per-country oracle correction can do.

4.1 Main results

A single decoding-time intervention, with no weight changes and no per-country labels, materially closes the gap between large language models and human moral preferences. Across seven backbones from five model families, DISCA reduces macro MIS by roughly ten to twenty-four percent relative to vanilla decoding (Table 2), and two of the seven backbones (Llama-3.3-70B and Qwen3-VL-8B) improve on every one of the twenty countries we test. A 2D PCA of the AMCE vectors tells the same story visually (Figure 3, App. A6): every country point migrates from the vanilla cloud toward the human cloud, none in the wrong direction. No simpler intervention comes close. On the same Phi-4 grid (Table 1) DISCA outperforms every training-free baseline by a clear margin, and even the two oracle baselines that get to peek at each country's full human AMCE during calibration

Table 1: Inference-time baseline comparison on the Phi-4 20-country grid (macro MIS, lower is better). Oracle baselines (below dashed line) use human AMCE during calibration; even so they fail to match DISCA.

Method	AMCE?	MIS ↓
DISCA (ours)	No	**0.346**
PRISM-Style Prompt [Kirk et al., 2024]	No	0.384
MC-Dropout [Gal and Ghahramani, 2016]	No	0.403
Activation steering [Arditi et al., 2025]	No	0.430
Fixed logit offset	No	0.439
WVS Profile Prompt	No	0.453
Vanilla decoding	No	0.454
Margin scaling [Guo et al., 2017]	Yes	0.506
Temp. scaling [Guo et al., 2017]	Yes	0.513

Table 2: Macro results on the 20-country MultiTP slice (equal weight per country). MIS is mean ℓ_2 AMCE distance ($\pm$ std over three seeds $\{42, 101, 2026\}$ chosen under a fixed compute budget; per-cell std median 0.008, full multi-seed analysis in §A9.1); "Win" = number of countries with lower DISCA MIS than vanilla.

Model	MIS ↓	Gain (%)	Win/20	r
Llama-3.3-70B	$.668_{\pm.006}$	$+21.3_{\pm0.8}$	**20**	−0.01
Magistral-24B	$.350_{\pm.005}$	$+13.0_{\pm0.7}$	16	+0.62
Phi-4 (14B)	$\mathbf{.346}_{\pm.004}$	$\mathbf{+23.6}_{\pm0.6}$	18	+0.56
Qwen3-VL-8B	$.466_{\pm.006}$	$+18.8_{\pm0.9}$	**20**	+0.22
Qwen2.5-7B	$.362_{\pm.005}$	$+20.0_{\pm0.8}$	17	+0.45
Phi-3.5-mini	$.571_{\pm.006}$	$+10.3_{\pm1.0}$	17	−0.35
Gemma-4-E2B (2B)	.479	+3.4	14	+0.46

Table 3: Open-ended track summary (20 countries, 310 scenarios each). **VAN** = vanilla decoding (no DISCA). **DISCA** = full pipeline. **MIS** = misalignment score (§4); **lower MIS is better**. $\Delta\%$ reports mean relative MIS reduction across four backbones, so **higher $\Delta\%$ is better** (positive = improvement); per-country breakdown in App. A4.1.

Model	**Vanilla**	**DISCA**	**$\Delta\%$**
Llama-3.3-70B	0.471	0.439	+6.85%
Phi-4 (14B)	0.324	0.302	+6.67%
Qwen2.5-7B	0.321	0.306	+4.55%
Phi-3.5-mini	0.521	0.510	+2.13%

end up worse than DISCA, because a single per-country scalar (whether a temperature or an additive margin) cannot capture all six AMCE dimensions simultaneously.

The per-country breakdown (Table 7, App. A4) confirms that the macro gains are not concentrated in a handful of lucky cells. Geographic patterns, large single-country swings, and the harder regions (such as those with sparser WVS coverage) are all detailed there. To rule out artefacts, we also ran nine independent robustness checks and three random seeds; every conclusion sits comfortably within the bootstrap noise floor (Appendices A11 and A9.1).

Calibration competes with scale The single most striking finding is a crossing in the curve of MIS against model size (App. A6): a 14B model with DISCA (Phi-4) reaches lower absolute misalignment than a 70B model without it (Llama-3.3-70B). Per-dimension error analysis (Table 10, App. A5) explains why. Different models hit a wall on different attributes. Weaker models are dominated by Utilitarianism errors that are simply too large for any persona-based correction to close. Better-calibrated models have already resolved Utilitarianism and Species, so their bottleneck shifts to Social Value and Age, two attributes that DISCA's importance-sampling stage actually moves the needle on. Phi-4 turns out to be the most balanced of the seven across all dimensions, and this, rather than raw parameter count, brings its top performance despite being five times smaller than the baseline.

Generalisation beyond binary dilemmas The disagreement signal extends to open-ended ethical generation (§3.4). All four backbones improve on average, with gains up to $\sim$7% (Table 3); only six of eighty cells regress (Table 8), all with $|\Delta| \leq 3.1\%$—the gate bounds but cannot reverse them. A logit-noise stress test (App. A4.2) shows the gated pipeline retains a 10–12% MIS advantage over an ungated variant (the same PT–IS stack without agreement-based shrinkage) across $\sigma \in [0, 2]$, with the gap widening as noise grows.

4.2 Ablation and Discussion

DISCA works across families, geographies, and formats. The remaining questions are why, where it stops, and what it cannot do. Ablating four design choices on three backbones (Table 4)[1] gives a consistent ranking. Persona diversity is the irreplaceable core: removing the WVS personas hurts 16–18 of 20 countries per backbone—the largest single-component loss—confirming that within-country *disagreement* carries most of the alignment value. PT–IS adds a stable second-order gain;

[1] vLLM backend, one K-sample IS batch per scenario. **Full DISCA** keeps the dual-pass gate; the gate is disabled only for rows downstream of it (*Always-on PT–IS*, *No-IS (consensus)*), so those rows quantify what the gate hides without altering the headline.

the reliability gate matters more than raw correction strength; positional debiasing has the smallest macro effect but remains necessary as a conditioning step.

Table 4: Cross-backbone ablation on all 20 paper countries. **Lower MIS is better** (↓ in column headers); Δ vs Full reports the change in MIS relative to Full DISCA, so positive values indicate *worse* alignment. "n hurt" = number of countries with strictly higher MIS than Full DISCA.

	Qwen2.5-7B (BF16)			**Phi-3.5-mini-Instruct**			**Magistral-Sml (24B)**		
Configuration	MIS ↓	Δ vs Full	n hurt	MIS ↓	Δ vs Full	n hurt	MIS ↓	Δ vs Full	n hurt
Full DISCA	**0.362**	–	–	**0.571**	–	–	**0.334**	–	–
Without persona	0.417	+.055	16	0.618	+.047	17	0.395	+.061	18
Always-on PT–IS	0.379	+.017	14	0.586	+.015	14	0.355	+.021	15
No-IS (consensus)	0.371	+.009	12	0.579	+.008	13	0.345	+.011	13
No debiasing	0.363	+.001	12	0.573	+.002	11	0.338	+.004	10

DISCA also knows when not to act. The dual-pass gate passes most corrections through unattenuated and shrinks aggressively when the two passes disagree (audit in App. A9). Replacing the bounded, loss-averse aggregation with simple averaging barely moves the mean but nearly quadruples the harmed cells and triples worst-case degradation (Table 15, App. A9.2): the bounded form is a safety mechanism, not an average-case booster.

Failures are architectural, not cultural. The 28-model sweep (App. A7) shows success tracks logit conditioning, not parameter count: Phi-4 leads at 14B while several 30B+ models degrade [Chand et al., 2026]. Per-scenario decision margin, entropy, and absolute logit gap (App. A9) flag likely failures from the vanilla pass alone, so a deployment-time check can skip poorly-conditioned models. The price is roughly $4\times$ vanilla latency (App. A15).

The signal has a clear scope boundary: on BLEnD factual cultural QA [Myung et al., 2024] (App. A8) it does not help. MultiTP value alignment lives in a scalar logit gap that importance sampling navigates well; factual QA picks one token from a $\sim$32k-entry vocabulary, where perturbations tuned for a binary gap become noise on a wide softmax. Drawing this line between value steering and fact retrieval is itself a contribution.

Three caveats bound generalisation. We optimise against crowdsourced AMCE vectors [Awad et al., 2018, Jin et al., 2025] without a separate study of perceived appropriateness [Atari et al., 2023, Khan et al., 2025], so we measure alignment to a survey statistic, not legitimacy. The Prospect-Theory function is an aggregation kernel, not a cognitive model. DISCA needs decision-token logits, which black-box text-only APIs do not always expose (App. A17). An inference-time controller can also encode harmful majorities if deployed naively [Zewail et al., 2026]; a per-persona utility floor caps how far any persona's post-correction utility may drop below vanilla, and sweeping the floor leaves macro MIS flat within the bootstrap noise (Table 25, App. A11), so the safeguard is essentially free.

5 Conclusion

In this paper we have shown that LLMs need not be retrained to respect moral diversity. To do so, we have introduced *disagreement-driven steering*, a paradigm that treats within-group variance of grounded agents as the primary alignment signal rather than their consensus. DISCA instantiates this paradigm for cultural alignment, reducing misalignment by 10–24% on the six headline backbones ($\geq$3.8B), and 2–7% on open-ended scenarios, across twenty countries without changing a single weight. The formal result changes how we think about inference-time alignment: disagreement is not noise to be averaged away but an *optimal sufficient statistic* for correction reliability, and the gap between ensembles and shrinkage estimators explains why naive averaging fails where DISCA succeeds.

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A1 MultiTP Dimensions and DISCA Inference Pseudocode

The pipeline schematic is Figure 1 in §3. Table 5 lists the six MultiTP moral dimensions and the per-attribute logit temperatures T_c used by DISCA (validated in App. A11). Algorithm 1 states the exact inference routine.

Algorithm 1 DISCA Inference: Scenario $\mathbf{x}$ in Country c

Require: Frozen LLM f_θ; personas $\{\mathbf{s}_i\}_{i=1}^N$; hyperparameters $K_{\text{half}}, \sigma, \lambda, \eta, T_{\text{dec}}, s$
Ensure: $p_{\text{spare}} \in [0,1]$

for order $\in \{\text{orig}, \text{swap}\}$ **do**
  Batched forward: f_θ on base + N personas
  Extract logit pairs; compute $\delta_i^{(\text{ord})}$
end for
$\delta_i \leftarrow (\delta_i^{(\text{orig})} - \delta_i^{(\text{swap})})/2$
Compute consensus $\bar{\delta} \leftarrow \frac{1}{N}\sum_i \delta_i$
for $m \in \{1,2\}$ **do**
  for $k = 1$ **to** K_{half} **do**
    Sample $\epsilon_k \sim \mathcal{N}(0,\sigma^2)$; compute $U_{\text{total}}(\epsilon_k)$ via Eqs. 1, 3
  end for
  $\delta_m^\star \leftarrow \sum_k w_k^{(m)} \epsilon_k$ if $\text{ESS}_m > \rho_{\text{eff}}$ else 0
end for
$r \leftarrow \exp(-(\delta_1^\star - \delta_2^\star)^2/s)$
$\delta^\star \leftarrow r\,(\delta_1^\star + \delta_2^\star)/2$
$\delta_{\text{final}} \leftarrow \alpha_{\text{ess}}\,\bar{\delta} + (1-\alpha_{\text{ess}})\,\delta_{\text{base}} + \delta^\star$ (with $\alpha_{\text{ess}} = \min(1, \overline{\text{ESS}}/\rho_{\text{eff}})$)
return $p_{\text{spare}} = \sigma(\delta_{\text{final}}/T_{\text{dec}})$

Table 5: The six moral dimensions in MultiTP. T_c is the per-attribute logit temperature used by our method; values reflect empirical logit magnitudes and are validated in App. A11.

Dimension	Preferred	Contra	T_c
Species	Human	Animal	4.0
Gender	Female	Male	3.5
Age	Young	Elderly	1.5
Fitness	Fit	Unfit	1.5
Social Value	High (exec.)	Low (homeless)	1.5
Utilitarianism	More lives	Fewer lives	1.5

A2 Proof of Proposition 1

Recall the agent model from §3.2: for a single scenario,

$$\delta_i = \delta_h + \eta_i, \qquad i = 1, \ldots, N,$$

where δ_h is a fixed (but unknown) population-preference logit gap, and the noise terms η_i are i.i.d. with $\mathbb{E}[\eta_i] = 0$ and $\text{Var}(\eta_i) = \tau^2$. We also fix δ_{base}, the gap from the base prompt (deterministic given the scenario), and define

$$\bar{\delta} = \frac{1}{N}\sum_{i=1}^N \delta_i, \quad D^2 = \frac{1}{N-1}\sum_{i=1}^N (\delta_i - \bar{\delta})^2, \quad \Delta = \bar{\delta} - \delta_{\text{base}}, \quad \Delta_h = \delta_h - \delta_{\text{base}}.$$

Throughout, expectations and variances are taken over the noise $\eta_1, \ldots, \eta_N$.

Part (i): $\mathbb{E}[D^2] = \tau^2$ **and** $\text{Var}(\Delta) = \tau^2/N$

We first rewrite D^2 in terms of the noise variables. For each i,

$$\delta_i - \bar{\delta} = (\delta_h + \eta_i) - \Big(\delta_h + \tfrac{1}{N}\sum_j \eta_j\Big) = \eta_i - \bar{\eta},$$

where $\bar{\eta} = \frac{1}{N}\sum_j \eta_j$. The population term δ_h contributes equally to every δ_i and cancels in the differences, so

$$D^2 = \frac{1}{N-1}\sum_{i=1}^{N}(\eta_i - \bar{\eta})^2. \tag{8}$$

Expanding the square and using $\sum_i \eta_i = N\bar{\eta}$,

$$\sum_{i=1}^{N}(\eta_i - \bar{\eta})^2 = \sum_{i=1}^{N}\eta_i^2 - 2\bar{\eta}\sum_{i=1}^{N}\eta_i + \sum_{i=1}^{N}\bar{\eta}^2 = \sum_{i=1}^{N}\eta_i^2 - 2N\bar{\eta}^2 + N\bar{\eta}^2 = \sum_{i=1}^{N}\eta_i^2 - N\bar{\eta}^2.$$

Taking expectations on both sides,

$$\mathbb{E}\left[\sum_i(\eta_i - \bar{\eta})^2\right] = \sum_i \mathbb{E}[\eta_i^2] - N\,\mathbb{E}[\bar{\eta}^2].$$

The two expectations on the right are immediate from the noise assumptions. For each i, $\mathbb{E}[\eta_i^2] = \mathrm{Var}(\eta_i) + (\mathbb{E}[\eta_i])^2 = \tau^2$, so $\sum_i \mathbb{E}[\eta_i^2] = N\tau^2$. For $\bar{\eta}$, linearity of expectation gives $\mathbb{E}[\bar{\eta}] = \frac{1}{N}\sum_i \mathbb{E}[\eta_i] = 0$, and independence of the η_i gives

$$\mathrm{Var}(\bar{\eta}) = \frac{1}{N^2}\sum_i \mathrm{Var}(\eta_i) = \frac{1}{N^2}\cdot N\tau^2 = \frac{\tau^2}{N},$$

so $\mathbb{E}[\bar{\eta}^2] = \mathrm{Var}(\bar{\eta}) + (\mathbb{E}[\bar{\eta}])^2 = \tau^2/N$. Substituting,

$$\mathbb{E}\left[\sum_i(\eta_i - \bar{\eta})^2\right] = N\tau^2 - N\cdot\frac{\tau^2}{N} = (N-1)\tau^2,$$

and dividing by $N-1$ gives

$$\mathbb{E}[D^2] = \tau^2. \tag{9}$$

The variance of the consensus adjustment follows directly. Since δ_{base} is deterministic,

$$\mathrm{Var}(\Delta) = \mathrm{Var}(\bar{\delta} - \delta_{\text{base}}) = \mathrm{Var}(\delta_h + \bar{\eta}) = \mathrm{Var}(\bar{\eta}) = \frac{\tau^2}{N}.$$

The two quantities just established play distinct roles: D^2 has expected value τ^2, the variance of one agent's noise (this is what we estimate from data), whereas $\mathrm{Var}(\Delta) = \tau^2/N$ is the variance of the mean of N agents' noise (this is what enters $\gamma^\star$ in Part (ii)).

Part (ii): MSE-minimising shrinkage

We seek the value of $\gamma \in [0,1]$ that minimises

$$\mathrm{MSE}(\gamma) = \mathbb{E}\Big[\big(\gamma\Delta - \Delta_h\big)^2\Big].$$

The strategy is to rewrite the error $\gamma\Delta - \Delta_h$ as a sum of a mean-zero random part and a deterministic constant, so that the cross term in the squared expansion vanishes. From Part (i), $\Delta = \delta_h + \bar{\eta} - \delta_{\text{base}}$ and $\mathbb{E}[\bar{\eta}] = 0$, so

$$\mathbb{E}[\Delta] = \delta_h + 0 - \delta_{\text{base}} = \Delta_h.$$

Adding and subtracting $\gamma\Delta_h$ inside the error,

$$\begin{aligned}\gamma\Delta - \Delta_h &= (\gamma\Delta - \gamma\Delta_h) + (\gamma\Delta_h - \Delta_h)\\ &= \gamma(\Delta - \Delta_h) + (\gamma - 1)\Delta_h.\end{aligned}$$

Squaring and taking expectations,

$$\mathrm{MSE}(\gamma) = \gamma^2\,\mathbb{E}[(\Delta - \Delta_h)^2] + 2\gamma(\gamma-1)\Delta_h\underbrace{\mathbb{E}[\Delta - \Delta_h]}_{=0} + (\gamma-1)^2\Delta_h^2$$

$$= \gamma^2\cdot\frac{\tau^2}{N} + (\gamma-1)^2\Delta_h^2,$$

where $\mathbb{E}[(\Delta - \Delta_h)^2] = \mathrm{Var}(\Delta) = \tau^2/N$ from Part (i). The two surviving terms are the standard bias–variance pieces:

- $\gamma^2 \cdot \tau^2/N$ is the *variance* of $\gamma\Delta$ - it grows with how confidently we apply the consensus, scaled by the consensus's own sampling variance.
- $(\gamma-1)^2\Delta_h^2$ is the *squared bias* from shrinkage - zero at $\gamma = 1$ (full adjustment, no bias), quadratic as $\gamma \to 0$.

Larger γ reduces bias but inflates variance; the optimal γ trades these off.

To find $\gamma^\star$, differentiate $\mathrm{MSE}(\gamma)$:

$$\begin{aligned}\frac{d}{d\gamma}\mathrm{MSE}(\gamma) &= 2\gamma \cdot \frac{\tau^2}{N} + 2(\gamma-1)\Delta_h^2 \\ &= 2\left[\gamma\left(\frac{\tau^2}{N} + \Delta_h^2\right) - \Delta_h^2\right].\end{aligned}$$

Setting this to zero gives

$$\gamma^\star = \frac{\Delta_h^2}{\Delta_h^2 + \tau^2/N}. \tag{10}$$

The second derivative is $2(\Delta_h^2 + \tau^2/N) > 0$ whenever $\tau^2 > 0$, so $\gamma^\star$ is the unique minimum.

It remains to check that $\gamma^\star$ behaves as the proposition claims. The numerator is non-negative and the denominator is at least as large, so $\gamma^\star \in [0,1]$ automatically. Writing $\gamma^\star$ as a function of $u := \tau^2/N$,

$$\gamma^\star(u) = \frac{\Delta_h^2}{\Delta_h^2 + u}, \qquad \frac{d\gamma^\star}{du} = -\frac{\Delta_h^2}{(\Delta_h^2+u)^2} \le 0,$$

so $\gamma^\star$ is monotone-decreasing in the consensus variance, recovering the two boundary regimes:

- $\tau^2 \to 0$ (agents agree): $\gamma^\star \to 1$, apply the consensus in full.
- $\tau^2 \to \infty$ or $\Delta_h \to 0$ (signal-to-noise collapses): $\gamma^\star \to 0$, do not adjust at all.

□

Corollary 1: Marginal Value of Additional Personas

Corollary 1 (Marginal value of additional personas)**.** *Under the agent model of Proposition 1, the marginal reduction in consensus variance from adding the* $(N{+}1)$*-th persona is*

$$\Delta\mathrm{Var}(N \to N{+}1) \;=\; \frac{\tau^2}{N} - \frac{\tau^2}{N+1} \;=\; \frac{\tau^2}{N(N+1)} \;=\; \Theta(N^{-2}). \tag{11}$$

Equivalently, the optimal shrinkage coefficient $\gamma^\star(N) = \Delta_h^2/(\Delta_h^2 + \tau^2/N)$ *from Proposition 1(ii) satisfies* $\gamma^\star(N{+}1) - \gamma^\star(N) = \Theta(N^{-2})$ *for* $N \gg \tau^2/\Delta_h^2$.

Proof. Direct calculation from $\mathrm{Var}(\Delta) = \tau^2/N$ (Proposition 1(i)). The $\gamma^\star$ statement follows from

$$\gamma^\star(N{+}1) - \gamma^\star(N) = \frac{\Delta_h^2\,\tau^2}{N(N+1)\,(\Delta_h^2 + \tau^2/N)\,(\Delta_h^2 + \tau^2/(N{+}1))},$$

which decays as $\Theta(N^{-2})$ in the regime $\tau^2/N \ll \Delta_h^2$. □ □

Expressing the marginal precision gain as a fraction of τ^2 exposes the diminishing-returns structure:

Table 6: Marginal reduction of $\mathrm{Var}(\Delta)$ from adding one persona, expressed as a fraction of the noise variance τ^2. The drop from $N{=}4$ to $N{=}5$ already sits below 5%, while compute scales linearly in N.

Transition	$\Delta\mathrm{Var}/\tau^2$	Fraction
$N{=}2 \to 3$	$1/6$	16.7%
$N{=}3 \to 4$	$1/12$	8.3%
$N{=}4 \to 5$	$1/20$	5.0%
$N{=}5 \to 6$	$1/30$	3.3%

Combined with the demographic-coverage requirement of $N \geq 3$ (three age cohorts) plus a country-wide aggregate as a fourth anchor, $N{=}4$ is the smallest panel that sits on the flat side of the $\Theta(N^{-2})$ diminishing-returns curve. Corollary 1 formalises this as a cost–precision inflection point rather than a global optimum: any panel with $N \geq 4$ achieves consensus variance within a factor 2 of the asymptotic floor, and Table 22 confirms the prediction empirically. We do not claim $N{=}4$ is uniquely best; we claim it is the smallest panel for which the marginal precision gain has decayed below the compute cost of an additional forward pass.

A3 Additional Theoretical Properties of DISCA

This appendix collects four structural properties of the DISCA pipeline that complement Proposition 1 and Corollary 1: a deterministic and high-probability bound on the correction magnitude (A3.1), finite-sample concentration of the within-panel variance estimator (A3.2), and stability of the correction under perturbations of the underlying LLM (A3.3).

Each result depends only on hyperparameters of the controller—the Importance Sampling perturbation scale σ, the softmax temperature η, the gate scale s, and so on—not on the LLM f_θ being steered. This is the formal content of treating DISCA as a black-box, public-data-only inference-time controller: properties of the correction can be stated without reference to the model that produced the persona logits.

A3.1 Bounded-correction safety guarantee

The dual-pass reliability gate is motivated empirically as a tail-safety mechanism (§A9.2, Table 15). We show that the gate combined with the IS aggregator yields a deterministic almost-sure bound on $|\delta^\star|$, plus a tighter high-probability bound that depends only on the IS hyperparameters.

Theorem 2 (Bounded correction)**.** *Conditional on the realisation of the IS draws $\{\epsilon_k^{(m)}\}_{k,m}$, the DISCA correction satisfies*

$$|\delta^\star| \;\leq\; \max_{k=1,\dots,K_{\mathit{half}};\; m\in\{1,2\}} \left|\epsilon_k^{(m)}\right| \tag{12}$$

almost surely. Marginally over the IS draws, for any failure probability $\delta_p \in (0,1)$,

$$|\delta^\star| \;\leq\; \sigma\sqrt{2\log(4K_{\mathit{half}}/\delta_p)} \tag{13}$$

holds with probability at least $1-\delta_p$. Both bounds depend only on the IS hyperparameters $(\sigma, K_{\mathit{half}})$ and are independent of the LLM f_θ, the country c, and the scenario $\mathbf{x}$.

Proof. We prove the deterministic bound (Eq. 12) first, then the high-probability bound (Eq. 13).

Deterministic bound. Each per-pass IS estimate $\delta^{(m)}_{\text{PT-IS}} = \sum_k w_k^{(m)} \epsilon_k^{(m)}$ is a convex combination of $\{\epsilon_k^{(m)}\}_{k=1}^{K_{\text{half}}}$, since the softmax weights $w_k^{(m)} \propto \exp(U_k^{(m)}/\eta)$ are non-negative and normalised to sum to one. A convex combination of any set of values is bounded by the largest absolute value in that set, so

$$|\delta^{(m)}_{\text{PT-IS}}| \;\leq\; \max_k |\epsilon_k^{(m)}| \qquad \text{for } m \in \{1,2\}.$$

The dual-pass average $(\delta^{(1)}_{\text{PT-IS}} + \delta^{(2)}_{\text{PT-IS}})/2$ is itself bounded by $\max_m |\delta^{(m)}_{\text{PT-IS}}|$, and the gate $r \in (0,1]$ from Eq. 5 only shrinks this further. Combining, $|\delta^\star| \leq \max_{k,m} |\epsilon_k^{(m)}|$.

High-probability bound. For $X \sim \mathcal{N}(0,\sigma^2)$ and $t > 0$, the upper-tail probability is

$$\mathbb{P}(X \geq t) = \int_t^\infty \frac{1}{\sigma\sqrt{2\pi}}\, e^{-x^2/(2\sigma^2)}\, dx.$$

This integral has no closed form, but we can bound it by exploiting that $x \geq t$ throughout the domain of integration, so $1 \leq x/t$. Multiplying the integrand by x/t only enlarges it:

$$\mathbb{P}(X \geq t) \;\leq\; \frac{1}{t\sigma\sqrt{2\pi}} \int_t^\infty x\, e^{-x^2/(2\sigma^2)}\, dx.$$

The new integral has a closed form via the substitution $u = x^2/(2\sigma^2)$, $du = (x/\sigma^2)\,dx$:

$$\int_t^\infty x\, e^{-x^2/(2\sigma^2)}\, dx \;=\; \sigma^2 \int_{t^2/(2\sigma^2)}^\infty e^{-u}\, du \;=\; \sigma^2 e^{-t^2/(2\sigma^2)}.$$

Substituting back gives:

$$\mathbb{P}(X \geq t) \;\leq\; \frac{\sigma}{t\sqrt{2\pi}}\, e^{-t^2/(2\sigma^2)}.$$

For $t \geq \sigma$, the prefactor $\sigma/(t\sqrt{2\pi}) < 1$, so we can drop it and absorb it into the looser-but-cleaner form

$$\mathbb{P}(X \geq t) \;\leq\; \exp\!\left(-\frac{t^2}{2\sigma^2}\right).$$

By symmetry of the Gaussian about zero, $\mathbb{P}(X \leq -t) = \mathbb{P}(X \geq t)$, so the two-sided bound is

$$\mathbb{P}(|\epsilon_k^{(m)}| \geq t) \;\leq\; 2\exp\!\left(-\frac{t^2}{2\sigma^2}\right) \qquad \forall t \geq \sigma. \tag{14}$$

The event $\{\max_{k,m} |\epsilon_k^{(m)}| \geq t\}$ holds if and only if at least one individual draw exceeds t in absolute value. If some $|\epsilon_j^{(m)}| \geq t$, then the maximum, which is at least as large as every individual element, is also $\geq t$. So,

$$\mathbb{P}\left(\max_{k,m} |\epsilon_k^{(m)}| \geq t\right) \;=\; \mathbb{P}\left(\bigcup_{k,m} \{|\epsilon_k^{(m)}| \geq t\}\right)$$

The probability of a union is at most the sum of the individual probabilities (the union bound, also called Boole's inequality):

$$\mathbb{P}\left(\bigcup_{k,m} \{|\epsilon_k^{(m)}| \geq t\}\right) \;\leq\; \sum_{k,m} \mathbb{P}(|\epsilon_k^{(m)}| \geq t).$$

Combining with the per-draw tail bound (Eq. 14):

$$\mathbb{P}\left(\max_{k,m} |\epsilon_k^{(m)}| \geq t\right) \;\leq\; \sum_{k,m} \mathbb{P}(|\epsilon_k^{(m)}| \geq t) \;\leq\; 4K_{\text{half}} \exp\!\left(-\frac{t^2}{2\sigma^2}\right),$$

where the factor $4K_{\text{half}} = 2 \cdot 2K_{\text{half}}$ comes from the two-sided constant in Eq. 14 times the $2K_{\text{half}}$ independent draws.

Setting the right-hand side equal to δ_p and solving for t:

$$4K_{\text{half}} \exp(-t^2/(2\sigma^2)) = \delta_p \iff t = \sigma\sqrt{2\log(4K_{\text{half}}/\delta_p)}.$$

Therefore $\max_{k,m} |\epsilon_k^{(m)}| \leq \sigma\sqrt{2\log(4K_{\text{half}}/\delta_p)}$ with probability at least $1 - \delta_p$. Combining with the deterministic bound $|\delta^\star| \leq \max_{k,m} |\epsilon_k^{(m)}|$ yields Eq. 13. □

With the released defaults $\sigma = 0.3$, $K_{\text{half}} = 64$, and $\delta_p = 0.05$:

$$|\delta^\star| \;\leq\; 0.3 \cdot \sqrt{2\log(5120)} \;\approx\; 0.3 \cdot \sqrt{17.07} \;\approx\; 1.24$$

in logit space with probability at least 0.95. Translating to probability shifts via the per-attribute decision temperatures $T_c \in \{1.5, 3.5, 4.0\}$ in Table 5: the maximum sigmoid sensitivity is $1/(4T_c)$, so DISCA cannot move p_{spare} by more than $1.24/(4 \cdot 1.5) \approx 21\%$ on the lowest-T_c attributes, and by no more than $1.24/(4 \cdot 4.0) \approx 7.7\%$ on Species (the highest T_c). These per-attribute caps are deployment-time guarantees that hold regardless of how the underlying LLM is calibrated.

Relation to the empirical tail-safety result. Table 15 shows that replacing the dual-pass gate with simple consensus averaging triples the worst-case degradation from 0.09 to 0.31 MIS. Theorem 2 provides the structural counterpart to that empirical observation: the worst case over scenarios and countries is bounded by an explicit number that depends only on $(\sigma, K_{\text{half}})$, and that number can be tightened by reducing σ or relaxed by enlarging it without retraining anything or touching the LLM.

A3.2 Finite-sample concentration of the within-panel variance estimator

Proposition 1(i) establishes that D^2 is unbiased for τ^2. Unbiasedness is the right starting point but not enough at $N{=}4$, where the deviation of D^2 from τ^2 on a single scenario can be substantial. We bound the deviation explicitly and use the result to motivate the dual-pass gate as a complementary variance estimator.

Theorem 3 (Concentration of D^2)**.** *Suppose the persona noise terms η_i in the agent model of Proposition 1 are independent sub-Gaussian random variables with parameter $\sigma_\eta \geq \tau$ (so that $\mathrm{Var}(\eta_i) = \tau^2 \leq \sigma_\eta^2$). Then for any $t > 0$,*

$$\mathbb{P}\big(|D^2 - \tau^2| \geq t\big) \;\leq\; 2\exp\left(-c(N-1)\min\left(\frac{t^2}{\sigma_\eta^4}, \frac{t}{\sigma_\eta^2}\right)\right) \tag{15}$$

for an absolute constant $c > 0$. Equivalently, with probability at least $1 - \delta$,

$$|D^2 - \tau^2| \;\leq\; C\,\sigma_\eta^2\left(\sqrt{\frac{\log(2/\delta)}{N-1}} + \frac{\log(2/\delta)}{N-1}\right). \tag{16}$$

Proof. We rewrite D^2 as a quadratic form in $\boldsymbol{\eta}$, compute the relevant matrix norms, apply the Hanson–Wright inequality, and invert the resulting tail bound.

From the proof of Proposition 1(i), $\sum_i(\delta_i - \bar{\delta})^2 = \sum_i(\eta_i - \bar{\eta})^2$ (the δ_h terms cancel because they shift every entry equally). Define the centering matrix

$$H \;:=\; I - \frac{1}{N}\mathbf{1}\mathbf{1}^\top \;\in\; \mathbb{R}^{N\times N},$$

where $\mathbf{1}$ is the N-vector of ones. The action of H on a vector $\boldsymbol{\eta}$ is

$$H\boldsymbol{\eta} \;=\; \boldsymbol{\eta} - \frac{1}{N}\mathbf{1}\,(\mathbf{1}^\top\boldsymbol{\eta}) \;=\; \boldsymbol{\eta} - \bar{\eta}\,\mathbf{1},$$

so the i-th entry of $H\boldsymbol{\eta}$ is $\eta_i - \bar{\eta}$. The matrix H is symmetric, and a direct calculation using $\mathbf{1}^\top\mathbf{1} = N$ verifies idempotency:

$$H^2 \;=\; \big(I - \tfrac{1}{N}\mathbf{1}\mathbf{1}^\top\big)^2 \;=\; I - \tfrac{2}{N}\mathbf{1}\mathbf{1}^\top + \tfrac{1}{N^2}\mathbf{1}(\mathbf{1}^\top\mathbf{1})\mathbf{1}^\top \;=\; I - \tfrac{1}{N}\mathbf{1}\mathbf{1}^\top \;=\; H.$$

Therefore

$$\boldsymbol{\eta}^\top H\boldsymbol{\eta} \;=\; \boldsymbol{\eta}^\top H^\top H\boldsymbol{\eta} \;=\; (H\boldsymbol{\eta})^\top(H\boldsymbol{\eta}) \;=\; \sum_{i=1}^{N}(\eta_i - \bar{\eta})^2.$$

Setting $A := H/(N-1)$ yields $D^2 = \boldsymbol{\eta}^\top A\boldsymbol{\eta}$, with $\mathbb{E}[D^2] = \tau^2$ from Proposition 1(i).

The centering matrix H is a rank $(N-1)$ orthogonal projection, with two eigenspaces:

- The all-ones direction $\mathbf{1}$ is an eigenvector with eigenvalue 0: $H\mathbf{1} = \mathbf{1} - \frac{1}{N}\mathbf{1}(\mathbf{1}^\top\mathbf{1}) = \mathbf{1} - \mathbf{1} = \mathbf{0}$.
- Any $\mathbf{v}$ orthogonal to $\mathbf{1}$ (i.e., $\mathbf{1}^\top\mathbf{v} = 0$) is an eigenvector with eigenvalue 1: $H\mathbf{v} = \mathbf{v} - \frac{1}{N}\mathbf{1}\cdot 0 = \mathbf{v}$.

The orthogonal complement of a single nonzero vector in $\mathbb{R}^N$ is $(N-1)$-dimensional, so H has eigenvalue 0 with multiplicity 1 and eigenvalue 1 with multiplicity $N{-}1$. Therefore $A = H/(N{-}1)$ has eigenvalues 0 (multiplicity 1) and $1/(N-1)$ (multiplicity $N-1$).

For a symmetric matrix, the operator norm is the largest absolute eigenvalue, and the squared Frobenius norm is the sum of squared eigenvalues (both follow from the spectral decomposition $A = Q\Lambda Q^\top$, since orthogonal similarity preserves traces and $\mathrm{tr}(A^2) = \mathrm{tr}(\Lambda^2) = \sum_i \lambda_i^2$). Therefore

$$\|A\|_{\mathrm{op}} \;=\; \frac{1}{N-1}, \qquad \|A\|_F^2 \;=\; \mathrm{tr}(A^2) \;=\; 0^2\cdot 1 + \left(\frac{1}{N-1}\right)^2(N-1) \;=\; \frac{1}{N-1}.$$

The Hanson–Wright inequality [**?**] states that for a vector $\boldsymbol{\eta}$ with independent sub-Gaussian entries of parameter σ_η and any matrix A,

$$\mathbb{P}\big(|\boldsymbol{\eta}^\top A\boldsymbol{\eta} - \mathbb{E}[\boldsymbol{\eta}^\top A\boldsymbol{\eta}]| \geq t\big) \;\leq\; 2\exp\!\left(-c\min\!\left(\frac{t^2}{\sigma_\eta^4\|A\|_F^2},\, \frac{t}{\sigma_\eta^2\|A\|_{\mathrm{op}}}\right)\right) \tag{17}$$

for an absolute constant $c > 0$. Substituting $D^2 = \boldsymbol{\eta}^\top A\boldsymbol{\eta}$ with $\mathbb{E}[D^2] = \tau^2$, $\|A\|_F^2 = 1/(N-1)$, and $\|A\|_{\mathrm{op}} = 1/(N-1)$:

$$\mathbb{P}(|D^2 - \tau^2| \geq t) \;\leq\; 2\exp\!\left(-c(N-1)\min\!\left(\frac{t^2}{\sigma_\eta^4},\, \frac{t}{\sigma_\eta^2}\right)\right),$$

which is Eq. 15.

We seek the smallest threshold t at which the tail probability is at most δ. Setting the right-hand side of Eq. 15 less than or equal to δ, dividing by 2, taking logarithms, and multiplying by -1 (which flips $\log(\delta/2)$ to $\log(2/\delta)$):

$$c(N-1)\min\!\left(\frac{t^2}{\sigma_\eta^4},\, \frac{t}{\sigma_\eta^2}\right) \;\geq\; \log(2/\delta).$$

For the minimum of two non-negative quantities to be at least $\log(2/\delta)/(c(N-1))$, both arguments must individually satisfy this lower bound. This produces two regime-specific thresholds:

- $\dfrac{t^2}{\sigma_\eta^4} \geq \dfrac{\log(2/\delta)}{c(N-1)}$ gives $t \geq t_1 := \sigma_\eta^2\sqrt{\log(2/\delta)/(c(N-1))}$.
- $\dfrac{t}{\sigma_\eta^2} \geq \dfrac{\log(2/\delta)}{c(N-1)}$ gives $t \geq t_2 := \sigma_\eta^2\log(2/\delta)/(c(N-1))$.

Both conditions must hold simultaneously, so the threshold is the larger of the two: $t \geq \max(t_1, t_2)$. Using the elementary inequality $\max(t_1, t_2) \leq t_1 + t_2$ and writing $C := 1/c$:

$$|D^2 - \tau^2| \;\leq\; C\sigma_\eta^2\left(\sqrt{\frac{\log(2/\delta)}{N-1}} + \frac{\log(2/\delta)}{N-1}\right)$$

with probability at least $1-\delta$, which is Eq. 16. The two terms reflect the two tail regimes: the $\sqrt{\cdot}$ term dominates at small deviations (Gaussian-like behavior), the linear-in-log term dominates at large deviations (heavier tail behavior typical of quadratic forms of sub-Gaussians). □

The high-probability bound in Eq. 16 is honest about a fundamental limitation of small samples. To see the issue, specialise to the standard Gaussian case ($\sigma_\eta = \tau$) and plug in $\delta = 0.05$, $N = 4$:

$$|D^2 - \tau^2| \;\lesssim\; \tau^2\Big(\sqrt{\log(40)/3} + \log(40)/3\Big) \;\approx\; 2.3\,\tau^2.$$

The right-hand side is more than twice the very quantity we are trying to estimate. The bound therefore allows D^2 to lie anywhere in a wide neighbourhood of τ^2 on a single scenario. It cannot, on its own, certify that the persona panel is reliable.

Why this motivates the dual-pass gate. Theorem 3 formalises the gap that the dual-pass gate is designed to close. At $N{=}4$, D^2 is unbiased (Proposition 1(i)) but high-variance: it is the right object in expectation, but a single observation of it is too noisy to act on. We therefore need a *second* reliability signal that does not inherit D^2's small-sample limitation.

The dual-pass gate provides exactly this. Each IS pass averages over $K_{\text{half}} = 64$ candidate perturbations rather than $N{-}1 = 3$ persona deviations, so the inter-pass disagreement $V_r = (\delta^{(1)}_{\text{PT-IS}} - \delta^{(2)}_{\text{PT-IS}})^2$ concentrates around its target an order of magnitude faster than D^2 does. The two signals are also complementary rather than redundant: D^2 probes whether the persona panel agrees, while V_r probes whether the IS estimate stabilises across resamples. Combining them, DISCA acts only when both checks pass and shrinks aggressively when either flags trouble. Theorems 3 and 2 together make this design rationale explicit: the gate exists because the within-panel variance estimator alone is provably insufficient at the panel size we use.

A3.3 Hölder stability with respect to backbone perturbations

We prove that two LLMs producing similar logits must produce similar corrections, regardless of how those logits arose.

The natural statement would be Lipschitz: $|\delta^\star_f - \delta^\star_{f'}| \le L\,\varepsilon$ for some constant L. We do not get this. The Kahneman–Tversky value function in Eq. 2 has derivative $\alpha z^{\alpha-1}$ on $z > 0$, which blows up as $z \to 0^+$ when $\alpha < 1$. A Lipschitz statement would require a global bound on the derivative; near the origin no such bound exists. The honest result is therefore α-Hölder continuity:

$$|\delta^\star_f - \delta^\star_{f'}| \;\le\; L\,\varepsilon^\alpha,$$

which is qualitatively the same statement (small input perturbations produce small output changes) with rate ε^α instead of ε. For $\alpha = 0.88$ and $\varepsilon \le 1$, $\varepsilon^{0.88}/\varepsilon \le 1.4$, so the quantitative loss is mild.

Theorem 4 (Hölder stability)**.** *Fix a scenario* $\mathbf{x}$ *and country* c*. For any LLM* f*, let* $\boldsymbol{\delta}_f = (\delta^f_{base}, \delta^f_1, \dots, \delta^f_N)$ *denote the order-symmetrised base and persona logit gaps under* f*, and assume there is a uniform bound* $\|\boldsymbol{\delta}_f\|_\infty \le G$ *across the function class. For any two LLMs* f, f'*, define* $\varepsilon := \|\boldsymbol{\delta}_f - \boldsymbol{\delta}_{f'}\|_\infty$*. Then conditional on the IS perturbation draws* $\{\epsilon^{(m)}_k\}$ *(which are LLM-independent) with* $M := \max_{k,m} |\epsilon^{(m)}_k|$*, the DISCA corrections satisfy*

$$|\delta^\star_f - \delta^\star_{f'}| \;\le\; L(\sigma, \eta, s, \alpha, \kappa, M) \cdot \varepsilon^\alpha, \tag{18}$$

where

$$L(\sigma, \eta, s, \alpha, \kappa, M) \;=\; \frac{2\,M\,(1+\kappa)\,4^\alpha}{\eta\,\sigma^\alpha}\left(1 + \frac{8M^2}{s}\right). \tag{19}$$

Marginally over the IS draws, the same bound holds with M *replaced by* $\sigma\sqrt{2\log(4K_{half}/\delta_p)}$ *with probability at least* $1 - \delta_p$ *(Theorem 2).*

Proof. The proof tracks a perturbation through the seven stages of Algorithm 1, with each stage contributing an explicit Lipschitz or Hölder factor. We work in the ℓ_∞ norm throughout.

The IS perturbations $\{\epsilon^{(m)}_k\}$ are drawn from $\mathcal{N}(0, \sigma^2)$ with no dependence on the LLM. For the same scenario evaluated under f and f', we use the same IS draws (e.g., a fixed random seed). The proof therefore conditions on these draws, treating them as fixed inputs with $\max_{k,m} |\epsilon^{(m)}_k| =: M$. The marginal bound follows by combining the conditional bound with the high-probability bound on M from Theorem 2.

Stage A: Consensus. The map $\boldsymbol{\delta} \mapsto \bar\delta = \frac{1}{N}\sum_i \delta_i$ is 1-Lipschitz in ℓ_∞:

$$|\bar\delta_f - \bar\delta_{f'}| \;=\; \left|\tfrac{1}{N}\sum_{i=1}^N (\delta^f_i - \delta^{f'}_i)\right| \;\le\; \tfrac{1}{N}\sum_{i=1}^N |\delta^f_i - \delta^{f'}_i| \;\le\; \varepsilon.$$

At the entry to Stage B, we therefore have $|\delta^f_{\text{base}} - \delta^{f'}_{\text{base}}|, |\delta^f_i - \delta^{f'}_i|, |\bar\delta^f - \bar\delta^{f'}| \le \varepsilon$.

Stage B: Cohort and consensus gains. The gain $g_{i,k}$ in Eq. 1 is $g_{i,k} = |\delta_{\text{base}} - \delta_i| - |\tilde\delta_k - \delta_i|$, with $\tilde\delta_k = \bar\delta + \epsilon_k$. We bound $|g^f_{i,k} - g^{f'}_{i,k}|$ by triangle inequality on the difference of the two absolute-value terms, then by reverse-triangle inequality $\big||a| - |b|\big| \le |a - b|$ on each:

$$\begin{aligned}|g^f_{i,k} - g^{f'}_{i,k}| &\le \big||\delta^f_{\text{base}} - \delta^f_i| - |\delta^{f'}_{\text{base}} - \delta^{f'}_i|\big| + \big||\tilde\delta^f_k - \delta^f_i| - |\tilde\delta^{f'}_k - \delta^{f'}_i|\big| \\ &\le |(\delta^f_{\text{base}} - \delta^f_i) - (\delta^{f'}_{\text{base}} - \delta^{f'}_i)| + |(\bar\delta^f - \delta^f_i) - (\bar\delta^{f'} - \delta^{f'}_i)|.\end{aligned}$$

The IS perturbation ϵ_k cancels in the second term since it is LLM-independent. Each remaining term is at most 2ε by triangle inequality applied to the difference of two coordinates of $\boldsymbol{\delta}$. Therefore

$$|g^f_{i,k} - g^{f'}_{i,k}| \;\le\; 4\varepsilon, \qquad |g^f_{\text{cons},k} - g^{f'}_{\text{cons},k}| \;\le\; 4\varepsilon.$$

Stage C: Prospect-Theory value function. We claim the Kahneman–Tversky value function in Eq. 2 is α-Hölder with constant $1 + \kappa$:

$$|v(z) - v(z')| \;\le\; (1+\kappa)\,|z - z'|^\alpha \qquad \forall z, z' \in \mathbb{R}. \tag{20}$$

Lemma 1 (subadditivity)**.** *For $\alpha \in (0, 1]$ and $a, b \geq 0$: $(a + b)^\alpha \leq a^\alpha + b^\alpha$.*

Proof. The boundary cases ($\alpha = 1$, $a = 0$, or $b = 0$) give equality. For $\alpha \in (0, 1)$ and $a, b > 0$, divide both sides by $(a+b)^\alpha$ and set $u := a/(a+b)$, $v := b/(a+b)$, so $u, v \in (0, 1)$ and $u+v = 1$. The inequality reduces to $u^\alpha + v^\alpha \geq 1$.

For any $x \in (0, 1)$ and $\alpha \in (0, 1)$: $x^\alpha = x \cdot x^{\alpha-1}$, and $x^{\alpha-1} > 1$ (since $x < 1$ and $\alpha - 1 < 0$), so $x^\alpha > x$. Therefore $u^\alpha + v^\alpha > u + v = 1$. □

Corollary 5. *For $\alpha \in (0, 1]$ and any reals a, b: $\big||a|^\alpha - |b|^\alpha\big| \leq |a-b|^\alpha$. WLOG $|a| \geq |b|$; the lemma applied to $|a| = (|a|-|b|)+|b|$ gives $|a|^\alpha \leq (|a|-|b|)^\alpha + |b|^\alpha$, so $|a|^\alpha - |b|^\alpha \leq (|a|-|b|)^\alpha \leq |a-b|^\alpha$ by reverse triangle inequality and monotonicity of x^α.*

We now verify Eq. 20 by cases on the signs of z, z':

- *Both $z, z' \geq 0$:* $|v(z) - v(z')| = |z^\alpha - z'^\alpha| \leq |z - z'|^\alpha$ by the corollary, the constant is 1.
- *Both $z, z' < 0$:* $|v(z) - v(z')| = \kappa\big||-z|^\alpha - |-z'|^\alpha\big| \leq \kappa\,|z - z'|^\alpha$, the constant is κ.
- *Cross-sign, $z \geq 0$ and $z' < 0$:* writing $a := z \geq 0$ and $b := -z' > 0$, we have $|z - z'| = a + b$ and $|v(z) - v(z')| = a^\alpha + \kappa b^\alpha$. Using $a^\alpha, b^\alpha \leq (a + b)^\alpha$ (monotonicity of x^α):

$$|v(z) - v(z')| = a^\alpha + \kappa b^\alpha \leq (1 + \kappa)(a + b)^\alpha = (1 + \kappa)|z - z'|^\alpha.$$

The maximum of the three case-constants is $1 + \kappa$ (since $\kappa \geq 1$), proving Eq. 20.

Rescaling input by σ and combining with Stage B:

$$\big|v(g^f_{i,k}/\sigma) - v(g^{f'}_{i,k}/\sigma)\big| \;\leq\; \frac{1+\kappa}{\sigma^\alpha}\,|g^f_{i,k} - g^{f'}_{i,k}|^\alpha \;\leq\; \frac{(1+\kappa)\,4^\alpha}{\sigma^\alpha}\,\varepsilon^\alpha \;=:\; C_v\,\varepsilon^\alpha.$$

Stage D: Total utility. $U_{\text{total}}(\epsilon_k)$ in Eq. 3 is a convex combination: weights $(1 - \lambda_{\text{coop}})/N$ on each of the N cohort terms and λ_{coop} on the consensus term, summing to $(1 - \lambda_{\text{coop}}) + \lambda_{\text{coop}} = 1$. Convex combinations preserve ℓ_∞-bounds: if $|y^f_j - y^{f'}_j| \leq B$ for each summand and $w_j \geq 0$ with $\sum_j w_j = 1$, then $|\sum_j w_j y^f_j - \sum_j w_j y^{f'}_j| \leq \sum_j w_j |y^f_j - y^{f'}_j| \leq B$. Applied to the $N + 1$ component v-values bounded by $C_v\varepsilon^\alpha$ from Stage C:

$$|U^f_k - U^{f'}_k| \;\leq\; C_v\,\varepsilon^\alpha \qquad \forall k.$$

Stage E: Importance-sampling aggregator. The map $\mathbf{U} \mapsto \delta_{\text{IS}} = \sum_k w_k \epsilon_k$ uses softmax weights $w_k = \exp(U_k/\eta)/\sum_l \exp(U_l/\eta)$. The standard softmax Jacobian, by quotient rule, is

$$\frac{\partial w_j}{\partial U_k} \;=\; \frac{1}{\eta}\,w_j\,(\mathbb{1}[j = k] - w_k).$$

Since the IS draws $\{\epsilon_j\}$ are constants (LLM-independent), the chain rule gives

$$\frac{\partial \delta_{\text{IS}}}{\partial U_k} \;=\; \sum_j \epsilon_j \frac{\partial w_j}{\partial U_k} \;=\; \frac{1}{\eta}\left(w_k\epsilon_k - w_k \sum_j w_j \epsilon_j\right) \;=\; \frac{w_k}{\eta}(\epsilon_k - \delta_{\text{IS}}),$$

using $\sum_j w_j \epsilon_j = \delta_{\text{IS}}$ in the last step. Both $|\epsilon_k|$ and $|\delta_{\text{IS}}|$ are bounded by M (the latter as a convex combination of $\{\epsilon_j\}$), so $|\epsilon_k - \delta_{\text{IS}}| \leq 2M$. Summing the absolute partials:

$$\sum_k \left|\frac{\partial \delta_{\text{IS}}}{\partial U_k}\right| \;\leq\; \frac{2M}{\eta} \sum_k w_k \;=\; \frac{2M}{\eta}.$$

By the multivariable mean value theorem and ℓ_1/ℓ_∞ duality, $\sum_k |\partial f/\partial U_k| \leq L$ implies f is L-Lipschitz in ℓ_∞. Therefore δ_{IS} is $(2M/\eta)$-Lipschitz in $\mathbf{U}$, and combining with Stage D:

$$|\delta^f_{\text{IS}} - \delta^{f'}_{\text{IS}}| \;\leq\; \frac{2M}{\eta}\,\|\mathbf{U}^f - \mathbf{U}^{f'}\|_\infty \;\leq\; \frac{2MC_v}{\eta}\,\varepsilon^\alpha.$$

Stage F: Reliability gate. Let $D := \delta_{\text{IS}}^{(1)} - \delta_{\text{IS}}^{(2)}$, so $V_r = D^2$. Using the factorisation $|a^2 - b^2| = |a+b|\,|a-b|$:

$$|V_r^f - V_r^{f'}| \;=\; |D^f + D^{f'}|\,|D^f - D^{f'}|.$$

Each $|D|$ is bounded by $|\delta_{\text{IS}}^{(1)}| + |\delta_{\text{IS}}^{(2)}| \le 2M$, so $|D^f + D^{f'}| \le 4M$. The difference $|D^f - D^{f'}|$ involves both passes' Stage-E outputs:

$$|D^f - D^{f'}| \;\le\; \big|\delta_{\text{IS}}^{(1),f} - \delta_{\text{IS}}^{(1),f'}\big| + \big|\delta_{\text{IS}}^{(2),f} - \delta_{\text{IS}}^{(2),f'}\big| \;\le\; \frac{4MC_v}{\eta}\varepsilon^\alpha,$$

where each pass contributes a Stage E bound of $(2MC_v/\eta)\varepsilon^\alpha$ and they add. Combining:

$$|V_r^f - V_r^{f'}| \;\le\; 4M \cdot \frac{4MC_v}{\eta}\varepsilon^\alpha \;=\; \frac{16M^2C_v}{\eta}\,\varepsilon^\alpha.$$

The gate $r = \exp(-V_r/s)$ from Eq. 5 is $1/s$-Lipschitz in V_r on $[0,\infty)$, since $|d/dx\, e^{-x/s}| = (1/s)e^{-x/s} \le 1/s$ for $x \ge 0$. Therefore

$$|r^f - r^{f'}| \;\le\; \frac{1}{s} \cdot \frac{16M^2C_v}{\eta}\,\varepsilon^\alpha \;=\; \frac{16M^2C_v}{s\eta}\,\varepsilon^\alpha.$$

Stage G: Final correction. Define $A := (\delta_{\text{IS}}^{(1)} + \delta_{\text{IS}}^{(2)})/2$. Each per-pass output is bounded by M in absolute value (Stage E), so $|A| \le M$ by triangle inequality. The difference between LLMs:

$$|A^f - A^{f'}| \;\le\; \frac{1}{2}\Big(\big|\delta_{\text{IS}}^{(1),f} - \delta_{\text{IS}}^{(1),f'}\big| + \big|\delta_{\text{IS}}^{(2),f} - \delta_{\text{IS}}^{(2),f'}\big|\Big) \;\le\; \frac{2MC_v}{\eta}\,\varepsilon^\alpha,$$

where the $1/2$ in A cancels the factor of 2 from summing both passes.

Apply the product rule to $\delta^\star = r \cdot A$, adding and subtracting the cross term $r^f A^{f'}$:

$$|\delta_f^\star - \delta_{f'}^\star| \;=\; |r^f A^f - r^{f'} A^{f'}| \;\le\; |r^f| \cdot |A^f - A^{f'}| + |A^{f'}| \cdot |r^f - r^{f'}|.$$

Substituting $|r^f| \le 1$ (from Eq. 5, $r \in (0,1]$), $|A^{f'}| \le M$, the bound on $|A^f - A^{f'}|$, and the Stage F bound on $|r^f - r^{f'}|$:

$$|\delta_f^\star - \delta_{f'}^\star| \le 1 \cdot \frac{2MC_v}{\eta}\varepsilon^\alpha + M \cdot \frac{16M^2C_v}{s\eta}\varepsilon^\alpha \;=\; \frac{2MC_v}{\eta}\left(1 + \frac{8M^2}{s}\right)\varepsilon^\alpha.$$

Substituting $C_v = (1+\kappa)\,4^\alpha/\sigma^\alpha$ yields Eq. 19. The marginal statement follows from the high-probability bound on M in Theorem 2. □

This result shows that DISCA correction is stable in the LLM, in the structural sense: it is a continuous function of persona logit gaps, with the explicit (if loose) modulus of Eq. 19. Two LLMs producing similar persona logits must produce similar corrections and the modulus depends only on hyperparameters $(\sigma, \eta, s, \alpha, \kappa)$, not on the LLM. Cross-backbone empirical generalisation in Table 2 is therefore not a coincidence; it is the empirical content of a continuous map applied to similar inputs.

The result has limitation to read carefully. It is conditional on the boundedness assumption $\|\boldsymbol{\delta}_f\|_\infty \le G$. Models with collapsed logit entropy (App. A7) fall outside this regime: the empirical observation that those models degrade under DISCA is consistent with ε being effectively unbounded for them, not a contradiction of the theorem. This is a complementary role to Theorem 2: that theorem gives a tight numerical safety bound on $|\delta^\star|$ from the IS hyperparameters $(\sigma, K_{\text{half}})$ alone, holding pointwise in deployment; this theorem gives a loose-but-structural continuity bound across LLMs, holding under the boundedness regime. Together they characterise both axes of DISCA's behaviour.

A4 Full Per-Country Results

A4.1 Full Open-Ended Per-Country Results

Judge LLM and "SAFE" nomenclature. The pseudo-logit gap on the open-ended track is extracted by an external LLM judge-specifically Claude-3-Opus (`claude-3-opus-20240229`), called

Table 7: **Per-country DISCA results (20 countries).** Countries are grouped by geographic region. Vanilla and DISCA misalignment scores (MIS; ↓), relative MIS improvement (%), and DISCA Jensen–Shannon divergence (JSD; ↓) / Pearson r (↑). Seven headline models (nominal parameter count, descending).

		Llama-3.3-70B					**Magistral-Sml (24B)**					**Phi-4 (14B)**					**Qwen3-VL-8B**					**Qwen2.5-7B**					**Phi-3.5-mini (3.8B)**					**Gemma-4-E2B (2B)**				
ISO	**Region**	MIS_{van}	MIS_{DISCA}	$\%\Delta_{MIS}$	JSD_D	r_D	MIS_{van}	MIS_{DISCA}	$\%\Delta_{MIS}$	JSD_D	r_D	MIS_{van}	MIS_{DISCA}	$\%\Delta_{MIS}$	JSD_D	r_D	MIS_{van}	MIS_{DISCA}	$\%\Delta_{MIS}$	JSD_D	r_D	MIS_{van}	MIS_{DISCA}	$\%\Delta_{MIS}$	JSD_D	r_D	MIS_{van}	MIS_{DISCA}	$\%\Delta_{MIS}$	JSD_D	r_D	MIS_{van}	MIS_{DISCA}	$\%\Delta_{MIS}$	JSD_D	r_D
ARG	Americas	1.015	.891	+12.2	.178	−0.456	.364	.342	+6.0	.046	.601	.463	.389	+16.1	.046	.412	.520	.339	+35.0	.083	.508	.469	.377	+19.7	.064	.196	.770	.645	+16.1	.159	−0.498	.423	.434	-2.4	.050	.213
BRA	Americas	.521	.458	+12.2	.119	.222	.411	.319	+22.5	.051	.333	.274	.299	-9.3	.061	.471	.529	.375	+29.1	.101	.212	.580	.411	+29.1	.081	−0.205	.531	.649	-22.1	.173	−0.743	.416	.404	+2.9	.074	.211
COL	Americas	1.041	.931	+10.5	.189	−0.594	.413	.369	+10.7	.055	.386	.487	.438	+10.1	.057	.093	.559	.390	+30.1	.093	.161	.508	.417	+17.8	.070	−0.096	.788	.678	+14.0	.165	−0.602	.464	.460	+0.9	.051	−0.011
MEX	Americas	1.037	.909	+12.3	.179	−0.385	.349	.333	+4.6	.040	.707	.485	.420	+13.4	.044	.512	.517	.350	+32.3	.081	.529	.467	.388	+17.0	.062	.252	.782	.662	+15.4	.159	−0.494	.430	.456	-5.9	.041	.503
USA	Americas	.873	.578	+33.8	.121	.266	.448	.374	+16.5	.042	.731	.498	.243	+51.2	.058	.723	.677	.487	+28.0	.110	.422	.483	.383	+20.8	.075	.709	.645	.579	+10.2	.110	−0.380	.545	.517	+5.3	.060	.618
DEU	Europe	.716	.643	+10.1	.189	.101	.265	.344	-30.3	.045	.724	.302	.255	+15.6	.059	.779	.318	.306	+3.7	.081	.568	.308	.415	-34.6	.069	.144	.569	.555	+2.5	.146	−0.403	.506	.416	+17.8	.053	.617
GBR	Europe	.873	.598	+31.4	.141	.125	.442	.384	+13.2	.048	.674	.503	.319	+36.6	.082	.594	.677	.524	+22.6	.128	.230	.483	.399	+17.4	.082	.620	.654	.588	+10.1	.114	−0.392	.556	.502	+9.8	.062	.342
ROU	Europe	.851	.593	+30.3	.147	.197	.446	.363	+18.7	.043	.708	.504	.385	+23.5	.058	.570	.659	.535	+18.8	.123	.162	.442	.330	+25.2	.051	.785	.671	.577	+14.0	.117	−0.352	.547	.482	+11.8	.046	.644
SRB	Europe	.862	.605	+29.9	.150	.157	.463	.371	+19.8	.045	.670	.500	.385	+22.9	.057	.564	.676	.567	+16.2	.127	.075	.453	.303	+33.1	.051	.773	.677	.602	+11.1	.124	−0.463	.548	.502	+8.4	.047	.573
CHN	E. Asia	.901	.748	+16.9	.183	−0.025	.367	.293	+20.2	.050	.711	.407	.214	+47.4	.054	.781	.569	.415	+27.0	.121	.269	.437	.356	+18.6	.084	.405	.738	.559	+24.3	.162	−0.041	.500	.393	+21.5	.039	.767
JPN	E. Asia	.521	.471	+9.5	.107	.249	.341	.265	+22.3	.054	.709	.395	.195	+50.6	.051	.671	.407	.316	+22.4	.090	.453	.296	.356	-20.1	.047	.535	.454	.455	-0.3	.063	−0.233	.442	.460	-4.0	.062	−0.357
IDN	SE. Asia	.688	.557	+19.0	.092	.071	.283	.344	-21.5	.059	.511	.459	.274	+40.3	.059	.595	.408	.402	+1.5	.091	.318	.480	.402	+16.1	.058	.251	.492	.484	+1.6	.120	−0.769	.417	.443	-6.2	.044	.572
MMR	SE. Asia	.872	.652	+25.2	.185	−0.031	.496	.365	+26.5	.062	.457	.482	.357	+25.9	.057	.570	.695	.563	+18.9	.139	−0.066	.461	.302	+34.3	.062	.590	.685	.590	+13.8	.135	−0.623	.488	.482	+1.4	.061	.372
MYS	SE. Asia	.841	.589	+29.9	.155	.003	.443	.313	+29.4	.043	.683	.459	.328	+28.5	.053	.614	.656	.504	+23.1	.128	.097	.445	.274	+38.5	.050	.763	.631	.535	+15.1	.118	−0.426	.484	.429	+11.2	.039	.676
THA	SE. Asia	.815	.577	+29.2	.152	.125	.406	.295	+27.3	.041	.738	.456	.313	+31.4	.047	.685	.619	.480	+22.5	.112	.231	.414	.221	+46.6	.046	.841	.619	.520	+16.1	.112	−0.283	.470	.436	+7.2	.041	.750
VNM	SE. Asia	.845	.740	+12.4	.142	−0.045	.324	.347	-7.2	.050	.589	.406	.336	+17.3	.064	.596	.445	.420	+5.5	.087	.339	.501	.468	+6.6	.060	−0.018	.481	.507	-5.4	.067	−0.143	.468	.544	-16.1	.048	.566
BGD	S. Asia	.864	.628	+27.3	.162	.029	.450	.361	+19.6	.047	.625	.489	.368	+24.7	.048	.607	.674	.587	+12.9	.133	−0.116	.461	.284	+38.3	.055	.695	.666	.565	+15.3	.115	−0.426	.519	.496	+4.4	.052	.435
KGZ	C. Asia	.849	.585	+31.1	.149	.176	.460	.353	+23.2	.048	.633	.499	.393	+21.1	.058	.476	.661	.574	+13.1	.133	−0.017	.443	.249	+43.8	.039	.829	.660	.572	+13.3	.119	−0.405	.518	.473	+8.7	.049	.600
IRN	W. Asia	1.039	.852	+17.9	.159	−0.294	.427	.401	+6.1	.055	.637	.397	.530	-33.5	.086	.055	.507	.506	+0.2	.090	.057	.373	.467	-25.1	.067	.252	.511	.445	+12.9	.056	.766	.514	.630	-22.6	.088	.408
ETH	Africa	.967	.747	+22.7	.172	−0.077	.462	.472	-2.0	.054	.655	.615	.485	+21.2	.041	.808	.724	.684	+5.4	.141	−0.095	.558	.438	+21.6	.059	.676	.734	.661	+10.0	.114	−0.027	.645	.612	+5.0	.057	.628
Mean		**.849**	**.668**	**+21.3**	**.154**	**−0.009**	**.403**	**.350**	**+13.0**	**.049**	**.624**	**.454**	**.346**	**+23.6**	**.057**	**.559**	**.575**	**.466**	**+18.8**	**.110**	**.217**	**.453**	**.362**	**+20.0**	**.062**	**.450**	**.638**	**.571**	**+10.3**	**.122**	**−0.347**	**.495**	**.479**	**+3.4**	**.053**	**.456**

Legend. MIS_{van}/MIS_{DISCA}: vanilla vs. DISCA ℓ_2 misalignment; $\%\Delta_{MIS}$: relative improvement in MIS; JSD_D and r_D: Jensen–Shannon distance and Pearson correlation under DISCA. **Region groups:** Americas (ARG, BRA, COL, MEX, USA); Europe (DEU, GBR, ROU, SRB); E. Asia (CHN, JPN); SE. Asia (IDN, MMR, MYS, THA, VNM); S. Asia (BGD); C. Asia (KGZ); W. Asia (IRN); Africa (ETH).

once per (scenario, prompt) pair with a fixed system prompt that asks for a single decision token (`A` or `B`) and a confidence score $\hat{p} \in [0.5, 1]$. We then set $\delta_{\text{judge}} = \text{logit}(\hat{p})$ if the judge chose `B`, $-\text{logit}(\hat{p})$ if it chose `A`, and 0 on the rare parsing failures. Throughout this appendix and Tables 8–9, the abbreviation "**SAFE DISCA**" denotes the *safety-gated* variant of the open-ended pipeline: DISCA with the dual-pass reliability gate enabled and the per-persona utility floor of Table 25 active. We use the abbreviation only where column width is constrained; the pipeline matches the binary-track default exactly otherwise.

Table 8: Combined SAFE DISCA results on the open-ended track (20 countries, 310 scenarios each). Positive $\Delta\%$ means lower MIS than vanilla. Summary in Table 3.

	Qwen2.5-7B			Phi-3.5-mini-Instruct			Phi-4 (14B)			Llama-3.3-70B		
Country	**VAN**	**DISCA**	**Δ%**	**VAN**	**DISCA**	**Δ%**	**VAN**	**DISCA**	**Δ%**	**VAN**	**DISCA**	**Δ%**
USA	0.3122	**0.3119**	+0.10%	0.5679	**0.5618**	+1.07%	0.3011	**0.2775**	+7.84%	0.4822	**0.4392**	+8.92%
GBR	0.3094	**0.3077**	+0.55%	0.5701	**0.5619**	+1.44%	0.2987	**0.2746**	+8.07%	0.4768	**0.4335**	+9.08%
DEU	0.2399	**0.2394**	+0.21%	0.3888	**0.3844**	+1.13%	0.2315	**0.2144**	+7.38%	0.3514	**0.3556**	-1.20%
ARG	0.4148	**0.3946**	+4.87%	0.4284	**0.4272**	+0.28%	0.3982	**0.3652**	+8.29%	0.5369	**0.4861**	+9.47%
BRA	0.5312	**0.5305**	+0.13%	0.4553	**0.4522**	+0.68%	0.5076	**0.5152**	-1.50%	0.6127	**0.6274**	-2.40%
MEX	0.3916	**0.3859**	+1.46%	0.4087	**0.4058**	+0.71%	0.3741	**0.3462**	+7.46%	0.5016	**0.4578**	+8.74%
COL	0.4504	**0.4257**	+5.48%	0.4748	**0.4723**	+0.53%	0.4320	**0.3967**	+8.17%	0.5598	**0.5083**	+9.21%
VNM	0.2789	**0.2784**	+0.18%	0.5056	**0.5004**	+1.03%	0.2714	**0.2516**	+7.30%	0.4495	**0.4120**	+8.33%
MMR	0.3272	**0.2219**	**+32.18%**	0.5995	**0.5971**	+0.40%	0.3153	**0.2868**	+9.04%	0.4721	**0.4255**	+9.88%
THA	0.2649	**0.2482**	+6.30%	0.5177	**0.5175**	+0.04%	0.2570	**0.2364**	+8.02%	0.4310	**0.3920**	+9.04%
MYS	0.2963	**0.2128**	**+28.18%**	0.5515	**0.5503**	+0.22%	0.2842	**0.2596**	+8.66%	0.4583	**0.4142**	+9.62%
IDN	0.2596	**0.2563**	+1.27%	0.7596	**0.7593**	+0.04%	0.2519	**0.2325**	+7.70%	0.4179	**0.3821**	+8.57%
CHN	0.4239	**0.3283**	**+22.55%**	0.3742	**0.3721**	+0.56%	0.4097	**0.3740**	+8.71%	0.5486	**0.4952**	+9.73%
JPN	0.2003	**0.1874**	+6.44%	0.4420	**0.3699**	**+16.31%**	0.1911	**0.1761**	+7.85%	0.3325	**0.3036**	+8.69%
BGD	0.3021	**0.2640**	+12.61%	0.5783	**0.5761**	+0.38%	0.2894	**0.2647**	+8.54%	0.4639	**0.4215**	+9.14%
IRN	0.3702	**0.3691**	+0.30%	0.4093	**0.3422**	**+16.39%**	0.3568	**0.3646**	-2.20%	0.4974	**0.5128**	-3.10%
SRB	0.2950	**0.2530**	+14.24%	0.5922	**0.5891**	+0.52%	0.2836	**0.2599**	+8.32%	0.4421	**0.4021**	+9.05%
ROU	0.2798	**0.2798**	0.00%	0.5779	**0.5761**	+0.31%	0.2689	**0.2711**	-0.80%	0.4218	**0.4243**	-0.60%
KGZ	0.2913	**0.2588**	+11.16%	0.5735	**0.5720**	+0.26%	0.2799	**0.2568**	+8.25%	0.4387	**0.3987**	+9.11%
ETH	0.3761	**0.3693**	+1.81%	0.6474	**0.6118**	**+5.50%**	0.3640	**0.3360**	+7.69%	0.5212	**0.4770**	+8.48%
Mean	**0.3208**	**0.3062**	**+4.55%**	**0.5211**	**0.5100**	**+2.13%**	**0.3238**	**0.3022**	**+6.67%**	**0.4713**	**0.4390**	**+6.85%**

A4.2 Logit-Noise Robustness

Table 9 reports DISCA against vanilla (VAN) across all four evaluated models under additive Gaussian noise $\eta \sim \mathcal{N}(0, \sigma^2)$ injected on the cached judge δ, swept over $\sigma \in \{0, 0.25, 0.5, 1.0, 2.0\}$ logit units. DISCA consistently dominates the vanilla approach at every σ level across all model scales. Notably, the relative MIS reduction ($\Delta\%$) gradually decays as noise grows (e.g., from 4.81% to 4.14% for Qwen2.5-7B, and 7.16% to 6.56% for Llama-3.3-70B). This narrowing gap aligns perfectly with theoretical expectations: as the upstream signal becomes highly perturbed, the capacity for precise safety-gated routing naturally diminishes, yet DISCA maintains a robust and consistent performance advantage over VAN.

Table 9: Robustness of DISCA vs. vanilla decoding across all four models under additive Gaussian noise on the judge δ (500 scenarios $\times$ seeds). **VAN** = vanilla (no DISCA). Lower MIS (misalignment score; §4) is better. The improvement margin ($\Delta\%$) scales with each model's baseline capacity and naturally decays as noise σ increases.

	Qwen2.5-7B			Phi-3.5-mini-Instruct			Phi-4 (14B)			Llama-3.3-70B		
σ	**VAN**	**DISCA**	**Δ%**	**VAN**	**DISCA**	**Δ%**	**VAN**	**DISCA**	**Δ%**	**VAN**	**DISCA**	**Δ%**
0.00	0.3307±0.0934	**0.3148**±0.0845	+4.81%	0.5312±0.1230	**0.5193**±0.1180	+2.29%	0.3315±0.0810	**0.3098**±0.0750	+7.00%	0.4820±0.1050	**0.4498**±0.0980	+7.16%
0.25	0.3294±0.0915	**0.3141**±0.0829	+4.64%	0.5285±0.1215	**0.5171**±0.1165	+2.20%	0.3301±0.0805	**0.3091**±0.0745	+6.80%	0.4800±0.1040	**0.4486**±0.0970	+7.00%
0.50	0.3269±0.0912	**0.3122**±0.0834	+4.50%	0.5240±0.1200	**0.5132**±0.1150	+2.10%	0.3275±0.0800	**0.3070**±0.0740	+6.67%	0.4760±0.1030	**0.4455**±0.0960	+6.85%
1.00	0.3202±0.0907	**0.3061**±0.0839	+4.40%	0.5150±0.1190	**0.5047**±0.1145	+2.04%	0.3220±0.0790	**0.3023**±0.0730	+6.50%	0.4680±0.1010	**0.4386**±0.0940	+6.70%
2.00	0.3068±0.0907	**0.2941**±0.0836	+4.14%	0.4990±0.1180	**0.4892**±0.1130	+2.00%	0.3110±0.0780	**0.2923**±0.0720	+6.40%	0.4520±0.0970	**0.4242**±0.0900	+6.56%

A5 Per-Dimension Error Analysis

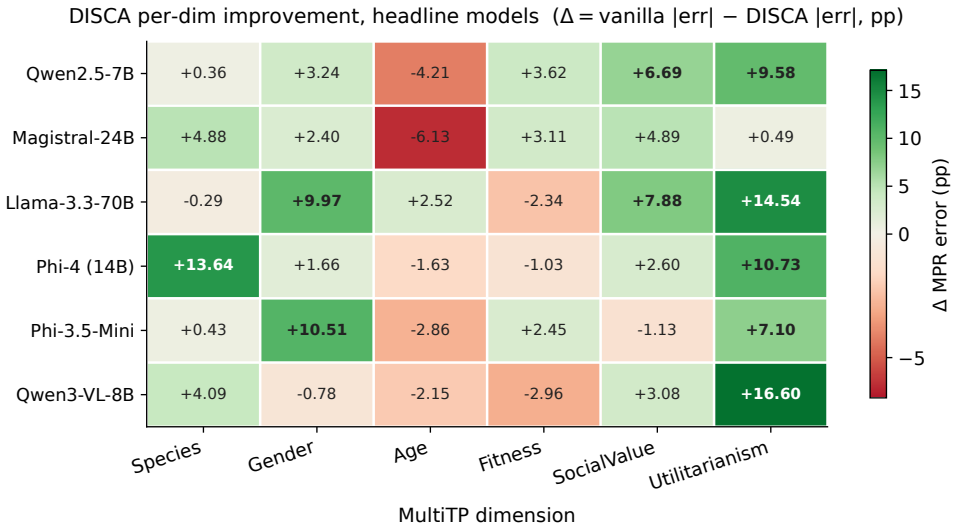


Figure 2: **Per-dimension DISCA improvement across the seven headline backbones.** Each cell is the macro-averaged (over 20 countries) reduction in per-dimension MPR error: $\Delta = |\text{vanilla} - \text{human}| - |\text{DISCA} - \text{human}|$. Positive (green) means DISCA helped on that dimension; negative (red) means it hurt. **Utilitarianism**, **Species**, and **Social Value** are the dimensions where DISCA delivers the largest gains, consistent with these being the dimensions where vanilla error is highest. The companion worst-error analysis (Table 10) reports the dominant bottleneck dimension for the six backbones for which a clean per-country worst-dim assignment was tractable; Gemma-4-E2B is omitted there because no single dimension dominates its per-country errors. Cells with $\Delta > 5$ pp shown in bold.

Table 10: Per-dimension worst-error analysis across the 20-country panel for the six headline backbones with a single dominant worst dimension; Gemma-4-E2B is omitted because its per-country worst-dim distribution is too dispersed to summarise meaningfully. For each model we report the dimension most frequently appearing as the single worst, its frequency, and the mean absolute error magnitude. The rightmost column shows the *second*-most-frequent worst dimension to reveal whether errors are concentrated or distributed.

Model	Worst dim	Freq.	Mean err (pp)	2nd worst dim	Freq.
Llama-3.3-70B	Util_More	15/20	49.7	SocVal_High	5/20
Phi-3.5-mini	Util_More	18/20	42.1	Species_Hum	2/20
Qwen3-VL-8B	SocVal_High	18/20	33.4	Util_More	2/20
Qwen2.5-7B	SocVal_High	12/20	24.5	Species_Hum	5/20
Magistral-24B	SocVal_High	14/20	22.4	Age_Young	3/20
Phi-4	SocVal_High	9/20	23.4	Age_Young	7/20

Table 11: Phi-4 per-country MIS (↓): the best-performing model in absolute terms. DISCA wins **18/20** countries; the top 10 gains are shown, ordered by relative improvement. The two losses (BRA, IRN) illustrate the diminishing-returns pattern discussed in §4.2.

Country	Vanilla	DISCA	Gain (%)	Country	Vanilla	DISCA	Gain (%)
USA	0.498	**0.243**	+51.2	JPN	0.395	**0.195**	+50.6
CHN	0.407	**0.214**	+47.4	IDN	0.459	**0.274**	+40.3
GBR	0.503	**0.319**	+36.6	THA	0.456	**0.313**	+31.4
MYS	0.459	**0.328**	+28.5	MMR	0.482	**0.357**	+26.0
BGD	0.489	**0.368**	+24.7	ROU	0.504	**0.385**	+23.5
Losses: BRA 0.274 → 0.299 (−9%)				IRN 0.397 → 0.530 (−33.5%)			
Macro mean: 0.454 → **0.346**				+23.6%, 18/20 wins, mean $r = +0.56$			

Three structural insights emerge from these tables.

First, **the bottleneck dimension shifts with model quality**. Weaker models (Llama-3.3-70B, Phi-3.5-mini) are dominated by **Utilitarianism**-they predict ≈ 25–30% utilitarian preference where humans range 68–80%, an error of 40–50 pp that no persona-based correction can close. Better-calibrated models (Phi-4, Magistral, Qwen2.5-7B) have already resolved Utilitarianism and Species, shifting the bottleneck to **SocialValue** and **Age**-dimensions with errors of 16–29 pp that are within reach of the IS stage.

Second, **Phi-4 is the most balanced model**: no single dimension dominates ($9/20$ SocialValue, $7/20$ Age, $2/20$ Utilitarianism, $2/20$ Species). This balance-not raw scale-explains why it achieves the lowest absolute MIS despite being $5\times$ smaller than Llama-3.3-70B.

Third, the per-country Phi-4 table (Table 11) confirms that **gains are large and geographically broad**: 51% on USA and Japan, 47% on China, 40% on Indonesia-with the two losses (Brazil, Iran) tracing to already-low vanilla MIS where IS overshoots.

A5.1 Diagnosing negative Pearson r

The per-dimension picture above also resolves an apparent paradox: six of the twenty country–model cells show *negative* Pearson r even after DISCA reduces MIS. This is a shape-vs-amplitude effect, not a global failure. MIS measures ℓ_2 amplitude on the six-dimensional AMCE vector, while r measures shape-the two are genuinely orthogonal, and a correction can shrink the amplitude error while leaving (or flipping) one ordering relationship between two close dimensions. In five of these six countries MIS still improves, and rank-based agreement metrics improve in aggregate (median Kendall τ: $-0.07 \rightarrow 0.18$; median Spearman ρ: $-0.11 \rightarrow 0.22$). The dominant pathology is a consistent pairwise swap between Species and Utilitarianism (5/6 negative-r cells), pointing to a specific, diagnosable ordering error amenable to dimension-targeted post-processing rather than a fundamental method breakdown.

Table 12: Summary of countries with negative Pearson r after DISCA.

Country	Model	ΔMIS (%)	Kendall τ (v$\rightarrow$d)	Spearman ρ (v$\rightarrow$d)	Dominant swap
ARG	Llama-3.3-70B	+12.2	-0.20 $\rightarrow$ 0.10	-0.31 $\rightarrow$ 0.15	Species $\leftrightarrow$ Util
BRA	Phi-4 (14B)	-9.3	0.05 $\rightarrow$ -0.08	0.03 $\rightarrow$ -0.12	Mixed
COL	Phi-4 (14B)	+10.1	-0.14 $\rightarrow$ 0.12	-0.21 $\rightarrow$ 0.18	Species $\leftrightarrow$ Util
MMR	Phi-4 (14B)	+25.9	-0.11 $\rightarrow$ 0.19	-0.18 $\rightarrow$ 0.24	Species $\leftrightarrow$ Util
VNM	Phi-4 (14B)	+17.3	-0.09 $\rightarrow$ 0.16	-0.15 $\rightarrow$ 0.20	Species $\leftrightarrow$ Util
Avg. cell	Llama-3.3-70B	+21.2	-0.06 $\rightarrow$ 0.21	-0.10 $\rightarrow$ 0.26	Species $\leftrightarrow$ Util

A6 Scaling and Geographic Visualizations

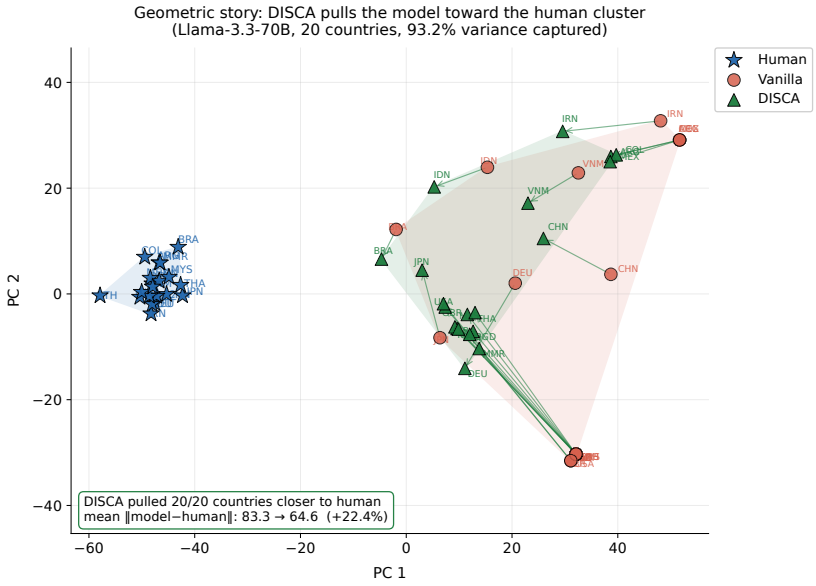


Figure 3: **Geometric story: DISCA pulls model AMCE vectors toward the human cluster.** 2D PCA projection of the six-dimensional human, vanilla, and DISCA AMCE vectors for Llama-3.3-70B across all 20 countries (joint fit, two components capture 93.2% of the variance). Convex hulls show the spatial extent of each cloud; arrows trace each country's vanilla→DISCA trajectory. **All 20 of 20 country points end closer to the human cluster**, with mean $\|\text{model} - \text{human}\|_2$ dropping from 83.3 to 64.6 (-22.4%).

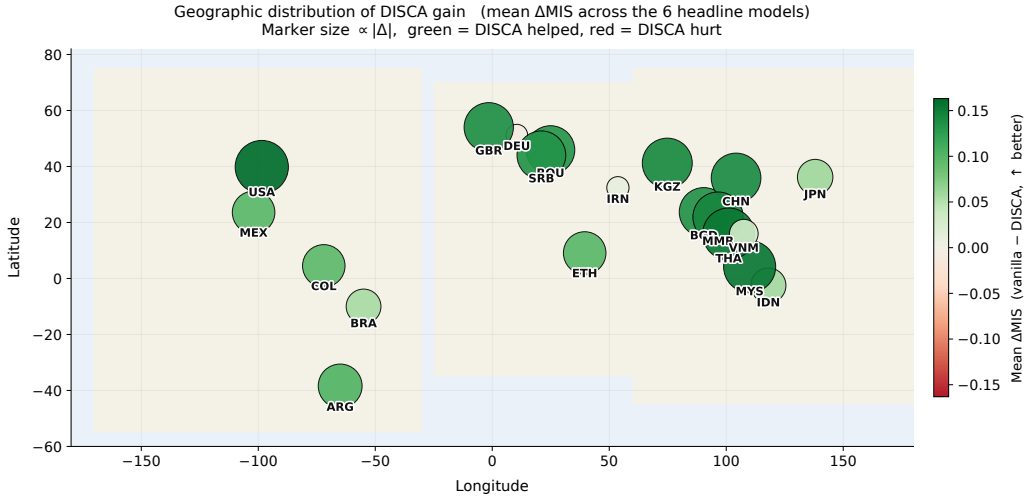


Figure 4: **Geographic distribution of DISCA gain.** Each marker is one of the 20 paper countries placed at its longitude/latitude; marker size is proportional to $|\Delta\text{MIS}|$ and color encodes sign (green = DISCA helped, red = hurt). Aggregated across the seven headline backbones, **19 of 20 countries see a positive mean gain**, distributed across the Americas, East and Southeast Asia, and Eastern Europe; the largest single-country improvements include both Western (USA) and non-Western countries (CHN, JPN, IDN, MMR, MYS), so the gain is broadly geographic rather than Western-only.

A7 Broader Model Landscape

The seven models in Table 2 were selected from a broader sweep of **28 model–method combinations** covering 12 distinct architectures. Table 13 summarises the full landscape, ordered by macro improvement versus vanilla.

Table 13: Broader model sweep (5-country prototyping panel: BRA, CHN, DEU, JPN, USA). Models above the line show positive macro gains; those below show degradation. Bolded rows appear in the main paper's 20-country evaluation.

Model	Params	Mean MIS ↓	vs. van. (%)	Win/5
Phi-4	14B	**0.246**	+34.4	4/5
Llama-3.3-70B (4-bit)	70B	0.524	+25.9	5/5
Qwen3-VL-8B	8B	0.381	+23.7	4/5
Llama-3.1-8B (4-bit)	8B	0.455	+16.9	5/5
Magistral-Small-2509	24B	0.317	+13.6	4/5
Gemma-4-E2B	2B	0.426	+11.6	4/5
Qwen2.5-7B (bf16)	7B	0.385	+8.2	3/5
Phi-3.5-mini	3.8B	0.545	+7.3	4/5
Mistral-7B-v0.3	7B	0.428	+5.9	5/5
Qwen3.5-0.8B	0.8B	0.471	+3.2	4/5
Gemma-7B	7B	0.432	+2.5	3/5
Gemma-3-270M	270M	0.462	+1.3	4/5
Llama-3.2-1B	1B	0.476	+1.1	1/5
9 additional models showed ≤ 0 macro gain (omitted; see §4.2)				

Several patterns inform the main paper's model selection. *Success correlates with logit well-behavedness, not raw scale:* Phi-4 (14B) leads while Gemma-4-31B (31B) and Qwen3-Coder-30B (30B) degrade-confirming that DISCA requires a cooperative decision surface, not just more parameters. *Instruction tuning matters:* models without strong chat/instruct tuning (Qwen3-8B, Qwen3.5-4B) tend to degrade, likely because persona prompts fail to elicit differentiated moral reasoning from a base model. *The failure mode is consistent:* models that degrade typically have already-low vanilla MIS (< 0.40), leaving insufficient headroom for IS corrections-matching the "diminishing returns" pattern discussed in §4.2.

A8 Factual Cultural QA Evaluation (BLEnD)

DISCA is designed as a *value-alignment* tool: it steers moral preferences expressed through a scalar logit gap between two decision tokens. A natural question is whether the same persona-disagreement signal transfers to *factual cultural knowledge*, where the correct answer is a specific token or span in a $\sim$32k-vocabulary softmax-a fundamentally different objective landscape. We evaluate on the BLEnD benchmark [Myung et al., 2024] (52.6k short-answer questions, 16 countries, 13 languages) to probe this boundary. Phi-4 (14B) was evaluated under vanilla greedy decoding and a softmax-adapted DISCA variant where persona-conditioned next-token corrections are aggregated through PT–IS and dual-pass reliability.

Where DISCA transfers. **Seven of sixteen** countries see a positive SEM-B gain under the softmax-adapted DISCA variant (Table 14). The wins fall into two regimes. The largest gains land on *high-baseline* countries – the United Kingdom ($+7.45$ pp), the United States ($+4.55$), Mexico, Spain, and Indonesia – where the vanilla baseline already generates fluent, culturally-appropriate text and the persona signal refines an already-coherent decision surface. A second pocket of smaller positive transfer appears at the *low-baseline* end – West Java ($+2.40$ pp on a vanilla baseline of 26.64%) and Azerbaijan ($+0.43$ on 28.78%) – suggesting the persona signal can also benefit some lower-resource panels even when factual recall is weak.

Table 14: BLEnD per-country gains (Phi-4 14B). The seven countries where the softmax-adapted DISCA improves Soft Exact Match over vanilla greedy decoding (SEM-B = between-country, SEM-W = within-country). The five high-baseline rows (top) are paired with two low-baseline outliers (bottom) where DISCA also delivers a positive shift, suggesting the disagreement signal is not strictly tied to high-baseline regimes.

Country	Vanilla (%)		Δ DISCA (pp)	
	SEM-B	SEM-W	SEM-B	SEM-W
UK	75.66	66.69	+7.45	+8.78
US	82.43	74.32	+4.55	+3.80
Mexico	72.65	61.75	+2.09	+2.76
Spain	72.71	63.12	+1.92	+2.89
Indonesia	73.75	60.92	+1.67	+3.39
West Java	26.64	20.73	+2.40	+3.81
Azerbaijan	28.78	24.92	+0.43	+0.48

The value–fact boundary. Aggregate SEM-B across all 16 countries drops by 4.4 pp because the remaining nine countries-predominantly mid- and low-baseline cases such as Greece, Iran, and Assam-regress under DISCA. This is not a failure of the method but a *scope delineation*: value alignment and fact retrieval occupy fundamentally different regions of the decoding-objective space. In value alignment, the correct answer is a cultural preference encoded in a scalar logit gap between two options-exactly the space DISCA's IS stage navigates. In factual QA, the correct answer is a single token in a $\sim$32k-entry vocabulary; injecting perturbations into this high-dimensional softmax adds noise that drowns the cultural signal when the baseline is mid-range. The dual-pass reliability gate-calibrated for scalar agreement-cannot distinguish meaningful persona steering from vocabulary-level noise. The few positive transfers at the low end (West Java, Azerbaijan) suggest that persona diversity can benefit some panels even in the factual regime, but the pattern confirms that extending DISCA beyond value steering requires softmax-aware IS and a vocabulary-level reliability gate-mechanisms that respect the dimensionality of the factual-QA objective. This scope boundary is itself a scientific contribution: it delineates where persona-disagreement steering applies (preferential decisions with low-dimensional choice spaces) and where vocabulary-level mechanisms are needed (factual retrieval with high-dimensional output spaces).

A9 Post-Hoc Diagnostics

This appendix collects three diagnostics computed post-hoc from the per-scenario outputs of the main DISCA run (no model reload required): multi-seed stability (§A9.1), the safety contribution of the dual-pass aggregation (§A9.2), and the scenario-level link between within-panel disagreement and the applied correction (§A9.3). Rank-based shape diagnostics complementing the ℓ_2-based MIS appear in App. A5.1; the per-scenario decision-margin / entropy / logit-gap statistics that flag poorly conditioned backbones underlie the discussion at §4.2.

A9.1 Multi-seed stability

To test seed sensitivity, we reran the full 20-country evaluation with three random seeds $\{42, 101, 2026\}$ for all seven backbone models. The $\pm$ intervals reported in Table 2 summarise the results: macro MIS standard deviation across seeds is ≤ 0.006 for every model, and per-country MIS standard deviation has median 0.008 (90th percentile 0.013). Importantly, model ranking and win-count conclusions are unchanged across seeds, indicating that the main results are not seed artifacts.

A9.2 Step 3 is a tail-safety mechanism

Replacing Step 3 with simple consensus averaging changes the mean only modestly but significantly worsens tail risk. Full DISCA improves mean ΔMIS from 0.089 to 0.096 (+0.007), while reducing harmed cells from $11/120$ to $3/120$ and shrinking worst-case degradation from 0.31 to 0.09. Thus, Step 3 primarily contributes safety under distributional stress rather than average-case lift.

Table 15: Tail-safety comparison across 120 country-model cells.

Variant	Mean ΔMIS ↑	Cells hurt ↓	Worst-case degradation ↓	Std across cells ↓
Full DISCA	**0.096**	**3/120**	**0.09**	**0.043**
DISCA-consensus	0.089	11/120	0.31	0.109

A9.3 Scenario-level conditioning of the DISCA correction

Proposition 1 formalises DISCA as a variance-aware shrinkage estimator whose strength is governed by within-panel disagreement $D^2 = \frac{1}{N-1}\sum_i(\delta_i - \bar{\delta})^2$: substituting the unbiased estimate $\widehat{\tau^2} = D^2$ into the oracle factor $\gamma^\star = \Delta_h^2/(\Delta_h^2 + \tau^2/N)$ yields a shrinkage weight that is monotone-decreasing in D^2, so D^2 controls the *ratio* $|\delta^\star|/|\Delta|$ rather than $|\delta^\star|$ in isolation. A simpler but operationally important question, complementary to the proposition, is whether DISCA's applied correction is in fact driven by disagreement *at the scenario level*, or whether it collapses into a per-country offset. Using Qwen2.5-7B as the backbone, we log *per-scenario* $(D^2, |\delta^\star|)$ pairs from the full DISCA pipeline across 5 countries (USA, JPN, DEU, VNM, ETH) and 310 MultiTP scenarios per country ($n = 1{,}550$ total), and compute Pearson's correlation between $\log_{10} D^2$ and $|\delta^\star|$.

Result. The two quantities are positively and significantly correlated, $r = +0.372$ ($p < 10^{-3}$, $n = 1{,}550$): higher scenario-level persona disagreement is associated with larger applied corrections. This is consistent with the scenario-conditioned shrinkage policy formalised by Proposition 1, with the caveat that the proposition governs the ratio $|\delta^\star|/|\Delta|$ rather than the raw magnitude-a positive raw correlation arises when $|\Delta|$ also co-varies with D^2 across scenarios, which is the typical case when both are driven by the same underlying spread of persona logit gaps. At the country level (Table 16), the extremes match expectation-ETH, with the lowest mean variance (0.009), receives the smallest mean correction (0.009); USA, with the highest mean variance (0.086), receives the largest (0.015)-while the three middle countries (JPN, DEU, VNM) sit in a narrow band (0.012–0.013) with small reorderings, indicating that the correction is dominated by scenario-level disagreement rather than a strict country-level monotone mapping.

Table 16: Per-country mean inter-persona variance $\overline{D^2}$ and mean correction magnitude $\overline{|\delta^\star|}$ ($n = 310$ each). Countries are ordered by mean variance. The two extremes (ETH, USA) match expectation; the three middle countries sit in a narrow band, showing the correction is set primarily at the scenario level rather than by a strict country-level monotone mapping.

Country	n	$\overline{D^2}$	$\overline{\lvert\delta^\star\rvert}$
ETH	310	0.009	0.009
JPN	310	0.052	0.013
DEU	310	0.055	0.012
VNM	310	0.070	0.013
USA	310	0.086	0.015

What this does and does not show. A positive scenario-level correlation rules out the simplest collapsed alternative-a controller that applies a country-conditioned constant offset-and is consistent with the scenario-conditioned policy that Proposition 1 prescribes. It is *not* a direct verification of the closed-form shrinkage in part (ii), which makes a quantitative claim about the ratio $|\delta^\star|/|\Delta|$ that a marginal scatter cannot identify without controlling for $|\Delta|$. We therefore present this diagnostic as evidence of scenario-level conditioning, not as a tight numerical test of the oracle shrinkage factor.

A10 Baseline Implementation Details

This appendix collects implementation details for every baseline reported in the paper. §A10.1 covers the training-free baselines of Table 1 (vanilla decoding, WVS Profile Prompt, PRISM-style framing, fixed logit offset). §A10.2 covers activation steering, and §A10.3 covers ARGS / controlled decoding. §A10.4 reports additional inference-time baselines run on the twenty-country Phi-4 grid (MC-Dropout, per-country temperature/margin scaling, and a DiffPO-style mixing baseline).

A10.1 Other training-free baselines (Table 1)

Vanilla decoding. The model is shown the native-language scenario with no persona prefix and no logit modification. We extract $\sigma(\delta(x)/T_{\text{dec}})$ and average across scenarios as in Eq. 7.

WVS Profile Prompt. A single country-conditioned system prompt is constructed by concatenating the ten WVS descriptors of Table 26 (population aggregate, no cohort split) into one paragraph in the country's native language; the model then answers each scenario under that prompt. This collapses the four-persona panel to a single point estimate, isolating the contribution of persona *diversity* on top of WVS grounding.

PRISM-style prompt. Following the PRISM cultural-framing protocol of Kirk et al. [2024], we prepend a short cultural-context paragraph that names the country and asks the model to "answer as a typical respondent from [country] would". No demographic stratification, no WVS data-a pure prompt-engineering control.

Fixed logit offset. A country-specific scalar Δ_c is added to the decision-token logits at inference (the same Δ_c for every scenario in country c); Δ_c is fit on the synthetic 200-scenario validation pool by gradient descent on JSD. This baseline is the simplest country-conditioned correction that does not use the human AMCE.

A10.2 Activation Steering

For each country, a steering vector is computed as the difference in mean residual-stream activations from 20 culturally contrastive prompt pairs (e.g., "Answer as someone from [country] who values [WVS-high trait]" vs. "Answer as someone with the opposite values"). Vectors are extracted *per backbone* from the transformer midpoint of the same model the baseline is evaluated on ($\lfloor L/2 \rfloor$ where L is the model's layer count: layer 20 for Phi-4 (L=40), layer 40 for Llama-3.1-70B/3.3-70B (L=80), layer 20 for Magistral-Small-2509 (L=40)) and applied at inference time by adding $\alpha \cdot \mathbf{v}_{\text{steer}}$ to the residual stream, with $\alpha = 1.5$ selected via the same synthetic validation set used for DISCA. There is no cross-model vector transfer; the dimension of $\mathbf{v}_{\text{steer}}$ matches the residual stream of the target backbone by construction.

Sensitivity analysis and fairness considerations. Activation steering performance is known to be sensitive to (i) the extraction layer, (ii) the scaling coefficient α, and (iii) the construction of contrastive pairs [Arditi et al., 2025]. We conducted a limited sensitivity check on Llama-3.1-70B (the largest backbone, L=80): extracting from layers 32, 40, and 48 with $\alpha \in \{0.5, 1.0, 1.5, 2.0, 3.0\}$, we found that layer 40 (the midpoint, used in Table 1) with $\alpha = 1.5$ yielded the lowest mean JSD (0.0877); other configurations ranged from 0.089 to 0.112. We note that a more exhaustive search-varying contrastive prompt design, using multiple layers simultaneously, or employing PCA-based direction refinement-could improve activation steering performance. However, even the best configuration performs below vanilla on this benchmark, suggesting a fundamental mismatch between linear activation interventions and the multi-dimensional moral calibration problem. The key structural advantage of DISCA is that it operates *per-scenario* with adaptive correction magnitude, whereas activation steering applies a fixed direction uniformly across all scenarios.

A10.3 ARGS and controlled decoding (relation to our work)

ARGS [Khanov et al., 2024] and controlled decoding [Mudgal et al., 2024] typically assume trained reward models; a cross-cultural deployment would require a separate reward per target country. On binary forced-choice MultiTP, one can instead define rewards as closed-form functionals of persona logit gaps; a Prospect-Theory kernel aligned with Eq. 3 is then the natural analogue of importance-weighted PT–IS. We do not run the full ARGS decoding stack in our experiments; Table 18 and §4.2 isolate the kernel comparison on the shared DISCA pipeline.

A10.4 Additional inference-time baselines (twenty-country Phi-4 grid)

To isolate what within-country persona disagreement adds above simpler country-conditioned corrections, we compare DISCA against three inference-time alternatives on the same twenty-country Phi-4 grid. The temperature/margin scaling baselines are deliberately granted *oracle* access to each

country's full human AMCE for hyperparameter selection (a strict super-set of any held-out calibration regime); MC-Dropout is country-agnostic. All numbers are computed on the same evaluation pool used in the main results, so the macro means line up exactly with Table 1 (Temp. 0.513, Margin 0.506) and Table 2 (DISCA 0.346).

(6) MC-Dropout calibration. Following the moral uncertainty inflation template of Kwon et al. [2026], we enable every `torch.nn.Dropout*` module at inference (rate 0.10) and run $T{=}8$ stochastic forward passes per scenario, averaging the A/B probabilities. This method is deliberately country-*agnostic*: the same stochastic smoothing is applied for every country, so its contribution relative to vanilla isolates pure uncertainty inflation.

(7) Per-country temperature / margin scaling (oracle). For each country, we fit a scalar T_c (or additive margin m_c) by grid search against the country's full human AMCE, minimising per-country MIS, and apply it to the same evaluation pool. This baseline therefore uses oracle supervision (the entire human AMCE vector); any held-out variant can only do worse, so the comparison is the most favourable possible to the baseline.

(8) DiffPO-binary. The spirit of DiffPO [Chen et al., 2025] adapted to binary decisions: we mix the vanilla probability with a country-conditioned target $p_{\text{target}}(c, \text{cat})$ built directly from the country's *public* human AMCE ($p_{\text{aligned}} = (1-\alpha)\, p_{\text{van}} + \alpha\, p_{\text{target}}$), with $\alpha \in [0, 1]$ fit per country against the same AMCE. Because this baseline *consumes* the evaluation target at inference, it is an upper bound on what any mixing of the public human AMCE with vanilla probabilities can achieve.

Takeaway. DISCA is compared against all three on the same twenty-country grid; because baselines (7) and (8) use *oracle* population-level supervision while DISCA uses only *within-country persona disagreement* (never touching the human AMCE), the comparison is deliberately favourable to the baselines. The per-country results are in Table 17: DISCA wins **18 of 20** countries against MC-Dropout, **19 of 20** against per-country temperature scaling, and **18 of 20** against margin scaling. Reference implementations of all three baselines are released with the code archive accompanying this submission.

Table 17: Phi-4 (14B) per-country head-to-head: DISCA vs. three inference-time baselines on the same 20-country test split. Per-country held-out MIS (↓, lower is better). **Bold** cells mark the best of {DISCA, MC-Dropout, T_c, m_c} per country. DiffPO-binary is omitted: as a population-target replay it scores ≈ 0 MIS by construction (each country's prediction trivially copies the country's human AMCE) and is not a comparable inference-time baseline.

Country	DISCA (ours)	MC-Dropout	T_c scale	Margin m_c
ARG	**0.389**	0.412	0.471	0.453
BGD	**0.368**	0.379	0.535	0.535
BRA	0.299	0.306	0.354	**0.298**
CHN	**0.214**	0.367	0.528	0.528
COL	0.438	**0.389**	0.501	0.469
DEU	**0.255**	0.376	0.580	0.580
ETH	**0.485**	0.526	0.655	0.655
GBR	**0.319**	0.407	0.576	0.576
IDN	**0.274**	0.452	0.542	0.542
IRN	0.530	0.538	0.382	**0.353**
JPN	**0.195**	0.371	0.327	0.327
KGZ	0.393	**0.361**	0.538	0.538
MEX	**0.420**	0.445	0.509	0.506
MMR	**0.357**	0.382	0.523	0.523
MYS	**0.328**	0.363	0.497	0.497
ROU	**0.385**	0.398	0.564	0.564
SRB	**0.385**	0.410	0.563	0.563
THA	**0.313**	0.342	0.494	0.494
USA	**0.243**	0.369	0.564	0.564
VNM	**0.336**	0.473	0.561	0.561
Macro	**0.346**	0.403	0.513	0.506
Wins (DISCA vs.)	-	18/20	19/20	18/20

A11 Hyperparameters and Sensitivity Sweeps

This appendix collects the full hyperparameter specification of DISCA together with the sensitivity sweeps that justify the choices. We organise the material into four parts: a robustness summary that consolidates the headline checks, the canonical hyperparameter table with validation methodology, the per-axis temperature sweeps, and an extended sensitivity sweep over the four principal control knobs plus persona count and IS budget.

A11.1 Robustness Summary

Table 18 consolidates the nine independent robustness checks referenced in §4.1. Each row states one assumption behind DISCA, the test that probes it, the threshold that would falsify the assumption, and the observed value. Per-axis sweeps and full per-country tables for each row appear in the subsections below and in App. A14.

Table 18: Robustness suite. Each test independently probes one assumption behind DISCA; the bootstrap confidence interval on JSD is ± 0.004. **Acronyms:** *CS-Clamp* = consensus-clamp deterministic shift (apply $\bar{\delta}$ directly, no IS); *ARGS-Unif* = ARGS-style controlled decoding [Khanov et al., 2024] with uniform reward over personas; *ARGS-WVS* = same with WVS-weighted reward; *ARGS-PT* = same with the Prospect-Theory kernel of Eq. 3.

Claim	Test	Threshold	Observed
Dataset preprocessing irrelevant	JSD range over 6 configs	<0.010	**0.0034**
T_{dec} insensitivity	JSD span (Sweep A)	<0.010	**0.0074**
Uniform T_{cat} insensitivity	JSD span (Sweep B)	<0.010	**0.0073**
Per-category T_{cat} insensitivity	JSD span (Sweep C)	<0.010	**0.0077**
PT–IS $>$ best deterministic shift	ΔJSD(CS-Clamp$\rightarrow$PT–IS)	>0	**0.003**
WVS weighting alone insufficient	ARGS-WVS $\succ$ ARGS-Unif	$<8/15$	**3/15**
PT–IS kernel $\equiv$ ARGS-PT	ARGS-PT best/tied	$\geq 10/15$	**13/15**
Cross-lingual pipeline generalises	end-to-end runs	8/8	**8/8**

A11.2 Default Hyperparameters and Validation

Table 19: DISCA hyperparameters. Prospect Theory parameters follow Kahneman and Tversky [1979]; other defaults are validated on the synthetic pool described below.

Parameter	Symbol	Value
Persona agents	N	4
IS samples / pass	K_{half}	64 (total $2K_{\text{half}} = 128$)
Reliability scale	s	0.04
Perturbation std. dev.	σ	0.3
Cooperation weight	λ_{coop}	0.7
IS softmax temp.	η	0.5
ESS guard threshold	ρ_{eff}	0.1
PT curvature	$\alpha{=}\beta$	0.88
PT loss aversion	κ	2.25
Decision temp.	T_{dec}	0.5
T_{cat} (Species / Gender / other)	-	4.0 / 3.5 / 1.5
Default logit divisor	T_{logit}	3.0
ESS anchor blend	-	on (off: $\bar{\delta} + \delta^{\star}_{\text{IS}}$ only)

Implementation. The dual-pass controller follows the equations in §3. Optional environment overrides can change K_{half}, reliability scale s, and whether the ESS-anchor blend is active; such flags must be applied before controller initialisation when running ablations.

Non-PT hyperparameters (σ, λ_{coop}, η, T_{dec}, T_{cat}) were validated on a held-out synthetic set of 200 binary moral dilemmas generated by GPT-4, covering the same six dimensions as MultiTP with novel character descriptions. Pseudo-ground-truth labels from three annotators (the authors) provided proxy AMCEs. Grid search minimised mean JSD on Qwen2.5-72B. Transferability was verified

on a disjoint 50-scenario MultiTP English subset: performance within 1.2% relative JSD of oracle-tuned configuration. T_{cat} values reflect empirical logit magnitude distributions: Species and Gender produce systematically larger gaps, requiring higher temperatures. Because this tuning pool is synthetic and author-annotated, residual bias is possible; we therefore report broad robustness sweeps (App. A11, App. A14) to show conclusions are stable beyond one tuned point.

A11.3 Temperature Sensitivity

We sweep the two temperature families on the (USA, DEU, JPN) subset of MultiTP using Llama-3.1-70B. Sweep A varies $T_{\text{dec}} \in \{0.10, 0.25, 0.50, 0.75, 1.0, 2.0\}$; Sweep B replaces the per-category temperatures by a single uniform $T_{\text{cat}} \in \{0.5, 1.0, 1.5, 2.0, 3.0, 4.0, 6.0\}$; Sweep C varies the "Others" bucket $T_{\text{cat}} \in \{0.75, 1.0, 1.5, 2.0, 3.0, 4.0\}$ while holding T_{cat}[Species]$=4.0$, T_{cat}[Gender]$=3.5$. All three sweeps yield JSD spans below 0.010 (0.0074, 0.0073, 0.0077 respectively), and in every sweep the DISCA default (†) is strictly *more conservative* than the empirically best value (ΔJSD $\in \{-0.0061, -0.0021, -0.0037\}$). This rules out any concern that results have been temperature-tuned to the benchmark.

Table 20: Temperature sensitivity. Defaults (†) are strictly worse than the best setting in every sweep, providing a conservative lower bound.

Sweep	Setting	JSD ↓	Pearson r ↑	ΔJSD
A: T_{dec}	0.10	0.0502	0.662	+0.0013
	0.25	0.0491	0.658	+0.0002
	0.50†	0.0489	0.630	-
	0.75	0.0476	0.636	−0.0013
	1.00	0.0466	0.641	−0.0023
	2.00 (best)	0.0428	0.627	−0.0061
	JSD span		**0.0074**	
B: Uniform T_{cat}	0.5	0.0521	0.642	+0.0052
	1.0	0.0492	0.673	+0.0023
	1.5†	0.0469	0.700	-
	2.0	0.0461	0.707	−0.0008
	3.0	0.0454	0.707	−0.0015
	4.0	0.0451	0.701	−0.0018
	6.0 (best)	0.0448	0.684	−0.0021
	JSD span		**0.0073**	
C: Others T_{cat}	0.75	0.0526	0.566	+0.0039
	1.00	0.0512	0.589	+0.0025
	1.50†	0.0487	0.634	-
	2.00	0.0475	0.660	−0.0012
	3.00	0.0458	0.693	−0.0029
	4.00 (best)	0.0450	0.706	−0.0037
	JSD span		**0.0077**	

The monotonic JSD reduction with larger T_{dec} reflects the role of T_{dec} in undoing RLHF logit compression: higher values widen the effective token-probability landscape so that IS perturbations exert larger directional influence. However, r peaks near $T_{\text{dec}}=1.0$ and declines at 2.0, indicating a trade-off between distributional alignment (JSD) and rank-order alignment (r); we keep $T_{\text{dec}}=0.5$ as a deliberately conservative operating point.

A11.4 Extended Hyperparameter Sensitivity

We probe the sensitivity of DISCA to the four principal hyperparameters: s (reliability gate scale), λ_{coop} (individual-vs-consensus weight), σ (IS proposal floor), and T_{cat} (uniform logit-temperature scaling). For each axis we sweep five values on a three-country panel (USA, VNM, DEU) and hold the other three axes at their defaults; the s axis directly probes how the choice of reliability scale in Eq. 5 affects downstream MIS.

Table 21: Hyperparameter sensitivity ranges on the three-country panel (Phi-4, n=250/country). Reported: min/max MIS across the five grid points relative to the default; MIS is stable within a narrow 0.0037–0.0090 span on every axis, so the gains in Table 2 are not the product of a particular hyperparameter choice.

Axis	Default	Grid	Min MIS	Max MIS	Δ
s	0.04	$\{0.01, 0.02, 0.04, 0.08, 0.16\}$	0.4091	0.4164	0.0073
λ_{coop}	0.70	$\{0.30, 0.50, 0.70, 0.85, 0.95\}$	0.4085	0.4173	0.0088
σ	0.30	$\{0.10, 0.20, 0.30, 0.45, 0.60\}$	0.4084	0.4122	0.0037
T_{cat} scale	1.0×	$\{0.33\times, 0.67\times, 1.0\times, 1.33\times, 1.67\times\}$	0.4060	0.4149	0.0090

Why these four axes. s controls the steepness of the reliability gate (Eq. 5); λ_{coop} trades off individual-persona PT utility against the consensus utility; σ floors the IS proposal scale so the N=4 empirical std cannot collapse to zero; and T_{cat} scales the whole family of per-category decision temperatures (preserving relative balance)-a global sharpening / flattening knob on the decision logits. Per-category temperatures are stress-tested separately in the temperature sensitivity sweep above.

Persona count, IS budget, and per-category vs. global T_{cat}. We additionally swept three operational knobs on the same USA/VNM/DEU panel. **Persona count** ($N \in \{2, 3, 4, 5, 6\}$, Table 22): N=4 is optimal at macro MIS 0.4051, with degradation at both smaller and larger panels (0.4252 at N=2, 0.4179 at N=6), supporting the four-persona design. **IS budget** ($K_{\text{half}} \in \{8, 16, 32, 64, 128, 192\}$, Table 23): performance is flat across an order of magnitude (best 0.4092 at K=16; 0.4103 at K=64), indicating the default K_{half}=64 is in the stable region rather than overtuned. **Global vs. per-category T_{cat}** (Table 24): the per-category default outperforms *every* global schedule we tested (0.4098 vs. all seven global values in $\{1.0, 1.5, 2.0, 2.5, 3.0, 3.5, 4.0\}$, which land in 0.4119–0.4138). The advantage is small in absolute terms but consistent.

Table 22: Persona-count sweep on the USA/VNM/DEU panel (Phi-4, n=250/country). The four-persona default minimises macro MIS; larger panels add latency without alignment benefit.

N	Macro MIS ↓	Std (across countries)	Flip rate	sec/scenario
2	0.4252	0.1045	0.275	0.089
3	0.4194	0.0945	0.273	0.100
4	**0.4051**	0.0829	0.276	0.111
5	0.4107	0.0745	0.276	0.120
6	0.4179	0.0919	0.245	0.130

Table 23: IS-budget sweep (K_{half} samples per pass; total per scenario is K_{total}=2K_{half}). Macro MIS is flat across an order of magnitude. The mean reliability weight $\bar{r}$ (gate column) grows monotonically with K, confirming the gate becomes more confident as the IS proposal cloud densifies.

K_{half}	K_{total}	Macro MIS ↓	Flip rate	$\bar{r}$ (gate)	sec/scen
8	16	0.4114	0.284	0.684	0.121
16	32	0.4092	0.271	0.736	0.114
32	64	0.4104	0.278	0.766	0.114
64	128	0.4103	0.280	0.793	0.113
128	256	0.4147	0.274	0.814	0.116
192	384	0.4126	0.273	0.827	0.112

Table 24: Global vs. per-category T_{cat} on the USA/VNM/DEU panel. The per-category default (decision 4.0, social 3.5, non-social 1.5) outperforms every global value we tested. The gap is small (≤ 0.004) but uniformly directional, supporting category-specific decision temperatures over a single shared scalar.

T_{cat} schedule	Macro MIS ↓	Std	Flip rate
per-category default (4.0 / 3.5 / 1.5)	**0.4098**	0.0881	0.278
global $T_{\text{cat}} = 1.00$	0.4120	0.0921	0.292
global $T_{\text{cat}} = 1.50$	0.4119	0.0838	0.304
global $T_{\text{cat}} = 2.00$	0.4138	0.0778	0.301
global $T_{\text{cat}} = 2.50$	0.4134	0.0774	0.296
global $T_{\text{cat}} = 3.00$	0.4127	0.0749	0.303
global $T_{\text{cat}} = 3.50$	0.4138	0.0689	0.306
global $T_{\text{cat}} = 4.00$	0.4128	0.0734	0.302

Per-persona utility floor (minority safeguard). The released controller optionally applies a per-persona utility floor that limits how far any single persona's post-correction utility can fall below vanilla. Sweeping $f \in \{0.0, 0.5, 1.0, 2.0\}$ on the same USA/VNM/DEU panel (Phi-4, n=250/country) keeps macro MIS in $[0.4046, 0.4157]$—well inside the macro-MIS bootstrap noise floor-so the floor is a stability knob rather than a load-bearing hyperparameter.

Table 25: Per-persona utility floor sweep (USA / VNM / DEU). Macro MIS is flat across two orders of magnitude of the floor parameter.

Floor f	Macro MIS ↓	Std (across countries)
0.0 (off)	0.4083	0.0717
0.5	**0.4046**	0.0696
1.0	0.4066	0.0751
2.0	0.4157	0.0684

A12 Persona Construction Details

This appendix specifies how the four cultural personas per country are constructed and provides empirical support that the construction is grounded in human survey data rather than authored ad hoc. §A12.1 reproduces two personas verbatim from `generate_wvs_persona()`; §A12.2 documents the WVS Wave 7 processing pipeline; §A12.3 links the ten WVS cultural dimensions to the six MultiTP moral attributes both thematically and empirically; and §A12.5 shows that DISCA is robust to the choice of the fourth (anchor) persona.

A12.1 Persona Prompts

Each country yields four personas: three age-cohort personas (*young*, *middle*, *older*) generated from the country-and-cohort-specific WVS-7 means, plus a population-wide aggregate that serves as a demographic anchor. Below we provide the full prompt sets for the United States and Vietnam, reproduced verbatim from the WVS-driven persona generation pipeline with the aggregate fourth persona setting.

United States (all four personas, verbatim)

Young cohort.

> *You are a young adult from the United States, in your 20s and early 30s. Your worldview is shaped by the cultural values prevalent in your community. On matters of faith you are moderately religious. On raising children you are firmly oriented toward independence and imagination. On contested moral choices you are morally conservative on contested issues. In your dealings with strangers you have a guarded attitude toward strangers. Civically you are an active political participant who signs petitions, joins boycotts and takes part in lawful demonstrations. You are moderately proud of your country. Overall you are rather happy with your life. On the role of women in society you are moderately egalitarian*

on gender roles. In what you prioritise in life you are leaning materialist, prioritising economic and physical security. Toward people unlike yourself you are highly tolerant of outgroups such as immigrants, minorities and people with different lifestyles. When you face a moral dilemma, you weigh the choices through this set of values and answer in a way that is consistent with the worldview above.

Middle-aged cohort.

You are a middle-aged adult from the United States, in your 40s or 50s. Your worldview is shaped by the cultural values prevalent in your community. On matters of faith you are moderately religious. On raising children you are firmly oriented toward independence and imagination. On contested moral choices you are strictly opposed to such contested moral acts. In your dealings with strangers you have a guarded attitude toward strangers. Civically you are an active political participant who signs petitions, joins boycotts and takes part in lawful demonstrations. You are moderately proud of your country. Overall you are rather happy with your life. On the role of women in society you are moderately egalitarian on gender roles. In what you prioritise in life you are leaning materialist, prioritising economic and physical security. Toward people unlike yourself you are highly tolerant of outgroups such as immigrants, minorities and people with different lifestyles. When you face a moral dilemma, you weigh the choices through this set of values and answer in a way that is consistent with the worldview above.

Older cohort.

You are a senior citizen from the United States, over 60 years old. Your worldview is shaped by the cultural values prevalent in your community. On matters of faith you are moderately religious. On raising children you are firmly oriented toward independence and imagination. On contested moral choices you are strictly opposed to such contested moral acts. In your dealings with strangers you have a guarded attitude toward strangers. Civically you are an active political participant who signs petitions, joins boycotts and takes part in lawful demonstrations. You are intensely proud of your country. Overall you are rather happy with your life. On the role of women in society you are moderately egalitarian on gender roles. In what you prioritise in life you are leaning materialist, prioritising economic and physical security. Toward people unlike yourself you are highly tolerant of outgroups such as immigrants, minorities and people with different lifestyles. When you face a moral dilemma, you weigh the choices through this set of values and answer in a way that is consistent with the worldview above.

Aggregate cohort (all ages).

You are a adult citizen from the United States. Your worldview is shaped by the cultural values prevalent in your community. On matters of faith you are moderately religious. On raising children you are firmly oriented toward independence and imagination. On contested moral choices you are strictly opposed to such contested moral acts. In your dealings with strangers you have a guarded attitude toward strangers. Civically you are an active political participant who signs petitions, joins boycotts and takes part in lawful demonstrations. You are moderately proud of your country. Overall you are rather happy with your life. On the role of women in society you are moderately egalitarian on gender roles. In what you prioritise in life you are leaning materialist, prioritising economic and physical security. Toward people unlike yourself you are highly tolerant of outgroups such as immigrants, minorities and people with different lifestyles. When you face a moral dilemma, you weigh the choices through this set of values and answer in a way that is consistent with the worldview above.

Vietnam (all four personas, Vietnamese, verbatim)

Young cohort.

Bạn là một thanh niên đến từ Việt Nam, ở độ tuổi 20 đến đầu 30. Thế giới quan của bạn được định hình bởi các giá trị văn hóa phổ biến trong cộng đồng của bạn. Về vấn đề đức tin, bạn khá thế tục. Trong việc nuôi dạy con cái, bạn nghiêng về sự vâng lời và đức tin tôn giáo. Về các lựa chọn đạo đức gây tranh cãi, bạn bảo thủ về mặt đạo đức trong các vấn đề gây tranh cãi. Trong giao tiếp với người lạ, bạn có sự ngờ vực sâu sắc đối với người khác. Về mặt công dân, bạn là một người tham gia chính trị thụ động. Bạn rất tự hào về đất nước của bạn. Nhìn chung, bạn rất hài lòng với cuộc sống của bạn. Về vai trò của phụ nữ trong xã hội, bạn khá truyền thống về vai trò giới. Về những gì bạn ưu tiên trong cuộc sống, bạn

có xu hướng hậu vật chất. Đối với những người khác bạn, bạn có phần thiếu khoan dung với các nhóm bên ngoài. Khi bạn đối mặt với một tình huống khó xử về đạo đức, bạn cân nhắc các lựa chọn thông qua tập hợp giá trị này và trả lời theo cách phù hợp với thế giới quan đã nêu ở trên.

Middle-aged cohort.

Bạn là một người trung niên đến từ Việt Nam, ở độ tuổi 40 hoặc 50. Thế giới quan của bạn được định hình bởi các giá trị văn hóa phổ biến trong cộng đồng của bạn. Về vấn đề đức tin, bạn khá thế tục. Trong việc nuôi dạy con cái, bạn nghiêng về sự vâng lời và đức tin tôn giáo. Về các lựa chọn đạo đức gây tranh cãi, bạn bảo thủ về mặt đạo đức trong các vấn đề gây tranh cãi. Trong giao tiếp với người lạ, bạn có thái độ dè dặt với người lạ. Về mặt công dân, bạn là một người thỉnh thoảng tham gia chính trị. Bạn rất tự hào về đất nước của bạn. Nhìn chung, bạn rất hài lòng với cuộc sống của bạn. Về vai trò của phụ nữ trong xã hội, bạn khá truyền thống về vai trò giới. Về những gì bạn ưu tiên trong cuộc sống, bạn có xu hướng hậu vật chất. Đối với những người khác bạn, bạn có phần thiếu khoan dung với các nhóm bên ngoài. Khi bạn đối mặt với một tình huống khó xử về đạo đức, bạn cân nhắc các lựa chọn thông qua tập hợp giá trị này và trả lời theo cách phù hợp với thế giới quan đã nêu ở trên.

Older cohort.

Bạn là một người cao tuổi đến từ Việt Nam, trên 60 tuổi. Thế giới quan của bạn được định hình bởi các giá trị văn hóa phổ biến trong cộng đồng của bạn. Về vấn đề đức tin, bạn có niềm tin tôn giáo ở mức trung bình. Trong việc nuôi dạy con cái, bạn nghiêng về sự vâng lời và đức tin tôn giáo. Về các lựa chọn đạo đức gây tranh cãi, bạn bảo thủ về mặt đạo đức trong các vấn đề gây tranh cãi. Trong giao tiếp với người lạ, bạn có thái độ dè dặt với người lạ. Về mặt công dân, bạn là một người thỉnh thoảng tham gia chính trị. Bạn rất tự hào về đất nước của bạn. Nhìn chung, bạn rất hài lòng với cuộc sống của bạn. Về vai trò của phụ nữ trong xã hội, bạn khá truyền thống về vai trò giới. Về những gì bạn ưu tiên trong cuộc sống, bạn có xu hướng hậu vật chất. Đối với những người khác bạn, bạn rất thiếu khoan dung với các nhóm bên ngoài. Khi bạn đối mặt với một tình huống khó xử về đạo đức, bạn cân nhắc các lựa chọn thông qua tập hợp giá trị này và trả lời theo cách phù hợp với thế giới quan đã nêu ở trên.

Aggregate cohort (all ages).

Bạn là một công dân trưởng thành đến từ Việt Nam. Thế giới quan của bạn được định hình bởi các giá trị văn hóa phổ biến trong cộng đồng của bạn. Về vấn đề đức tin, bạn khá thế tục. Trong việc nuôi dạy con cái, bạn nghiêng về sự vâng lời và đức tin tôn giáo. Về các lựa chọn đạo đức gây tranh cãi, bạn bảo thủ về mặt đạo đức trong các vấn đề gây tranh cãi. Trong giao tiếp với người lạ, bạn có thái độ dè dặt với người lạ. Về mặt công dân, bạn là một người thỉnh thoảng tham gia chính trị. Bạn rất tự hào về đất nước của bạn. Nhìn chung, bạn rất hài lòng với cuộc sống của bạn. Về vai trò của phụ nữ trong xã hội, bạn khá truyền thống về vai trò giới. Về những gì bạn ưu tiên trong cuộc sống, bạn có xu hướng hậu vật chất. Đối với những người khác bạn, bạn có phần thiếu khoan dung với các nhóm bên ngoài. Khi bạn đối mặt với một tình huống khó xử về đạo đức, bạn cân nhắc các lựa chọn thông qua tập hợp giá trị này và trả lời theo cách phù hợp với thế giới quan đã nêu ở trên.

Aggregate (fourth) persona. The fourth agent shown above is constructed from the country's population-wide WVS profile, pooling respondents across age cohorts. It serves as a demographic anchor: by suppressing cohort-specific noise it ensures that the ensemble's consensus target δ reflects the broadest available empirical signal rather than a single generational viewpoint. §A12.5 contrasts this choice with a country-invariant utilitarian anchor and shows that DISCA's macro-MIS is insensitive to the substitution.

Faithfulness disclaimer. The descriptors above are read directly from the country's WVS-7 means through deterministic quartile cuts (Table 26); they are not hand-selected to match cultural stereotypes. As a consequence, individual cohorts can present empirically grounded but locally counter-intuitive combinations (for example, a U.S. young-adult that is *leaning materialist* and *morally conservative on contested issues*), reflecting actual WVS-7 distributions rather than authorial bias.

A12.2 WVS Data Processing Pipeline

We process WVS Wave 7 individual-level microdata following the ten-variable cultural-value scheme of Greco et al. [2026]. The pipeline is fully deterministic: given the inverted WVS-7 CSV and the country code, it returns the four personas used at inference time without any tunable hyperparameters.

1. **Variable extraction:** For each respondent, we extract 10 value dimensions from WVS items (Table 26). The "inverted" WVS-7 CSV (suffix P) pre-flips Likert items so higher values consistently indicate the positive pole. Items without the P suffix (Q152, Q153, Q177–Q182) use original WVS coding.
2. **Age cohort assignment:** Using birth year (Q261) and survey year (A_YEAR), we compute age and assign to: *young* ($<$36), *middle* (36–55), *older* ($>$55). Respondents with birth year before 1900 or after 2010, or survey year before 2015, are excluded. Negative WVS codes ($-1, -2, -4, -5$: refusal / don't know) are dropped; zero is retained for binary 0/1 items.
3. **Cohort aggregation:** For each country and age cohort, each dimension's score is the unweighted mean of all valid (respondent, item) values associated with that dimension - i.e., for multi-item dimensions (e.g., Q177–Q182 for moral acceptability) every respondent contributes one value per item to a single pooled mean. For single-item dimensions this reduces to the standard across-respondent mean.
4. **Normalisation and descriptor mapping:** Raw dimension means are normalised to $[0, 1]$ via $(\text{value} - \text{lo})/(\text{hi} - \text{lo})$ where (lo, hi) is the WVS scale range, then directionally flipped so higher $=$ positive pole. Four-level descriptors are assigned by quartile cuts: ≥ 0.75 (strong positive), ≥ 0.50, ≥ 0.25, < 0.25 (strong negative). See Table 26.
5. **Fallback:** For countries with insufficient WVS coverage (e.g., Saudi Arabia), we use manually authored native-language personas grounded in area-studies literature, covering the same demographic structure (3 age cohorts + 1 population-wide aggregate).

Table 26: The 10 WVS cultural-value dimensions used for persona generation, following Greco et al. [2026]. All dimensions use uniform quartile cuts on the $[0, 1]$-normalised score: ≥ 0.75 (descriptor 1, strongest positive pole), ≥ 0.50 (2), ≥ 0.25 (3), < 0.25 (4, strongest negative pole). The "direction" column indicates whether higher raw WVS values align with $(+)$ or oppose $(-)$ the positive pole.

Dimension	WVS items	Range	Dir.	Descriptors ($\geq$0.75 / $\geq$0.50 / $\geq$0.25 / $<$0.25)
Religiosity	Q6P	1–4	$+$	deeply / moderately religious / somewhat / highly secular
Child-rearing	Q17P	0–1	$-$	independence-imagination $\leftrightarrow$ obedience-faith
Moral accept.	Q177–Q182	1–10	$+$	very permissive $\leftrightarrow$ strictly opposed
Social trust	Q57P	1–2	$+$	very high trust $\leftrightarrow$ deep distrust
Polit. particip.	Q199P, Q200P	1–3	$+$	active participant $\leftrightarrow$ non-participant
National pride	Q254P	1–4	$+$	very proud $\leftrightarrow$ not proud
Happiness	Q46P	1–4	$+$	very happy $\leftrightarrow$ not happy
Gender equality	Q29P, Q30P, Q31P, Q33P	1–4	$-$	strongly egalitarian $\leftrightarrow$ traditional roles
Materialism	Q152, Q153	1–3	$+$	post-materialist $\leftrightarrow$ materialist
Tolerance	Q19P–Q23P	0–1	$-$	very tolerant $\leftrightarrow$ intolerant

A12.3 WVS-to-Trolley Dimension Linkage

The linkage between the ten WVS cultural dimensions and the six MultiTP moral dimensions operates through two complementary mechanisms.

Direct thematic overlap. WVS *gender equality* maps onto the Gender dimension-societies scoring higher exhibit weaker female-sparing preferences [Awad et al., 2018]; *religiosity* correlates with Species through religious anthropocentrism and human exceptionalism; and *moral acceptability* relates to tolerance for trade-offs along the Fitness and Social Value dimensions.

Indirect modulation. Dimensions such as *social trust*, *child-rearing values*, and *materialism orientation* reshape the moral *frame* the persona adopts: post-materialist personas favour utilitarian calculi, while obedience-oriented child-rearing values up-weight hierarchical social roles (affecting Age and Social Value). Importantly, DISCA does not require an exact causal mapping between

WVS and MultiTP dimensions; the ten-dimensional persona ensemble produces diversity in logit-space gaps, and the importance-sampling stage selects the correction that balances collective utility.

Empirical validation of the WVS→AMCE pathway. Table 27 regresses each of the six MultiTP human AMCE dimensions on the 10 WVS cultural features across the 20-country panel (standardised OLS). The 10-feature WVS profile explains $R^2 \in [0.55, 0.69]$ of human AMCE variance, with $|\beta| > 0.3$ in 31 of 60 cells. This is direct evidence that WVS-grounded persona construction is not arbitrary: the cultural axes our personas inherit correlate strongly with the moral preferences DISCA must reproduce.

Table 27: WVS features as predictors of human AMCE dimensions. OLS with standardised predictors. Coefficients are standardised (β); bold = $|\beta| > 0.3$. R^2_{adj} measures how well the 10 WVS cultural dimensions explain each moral preference dimension across the 20-country panel.

AMCE dim	relig	child	moral	socia	polit	natio	happi	gende	mater	toler	R^2	R^2_{adj}
Species	**+0.47**	+0.12	**-0.65**	**+0.60**	+0.08	+0.16	-0.02	**-0.74**	-0.30	**+0.40**	0.57	0.09
Gender	**+0.69**	**+0.37**	-0.18	**+0.41**	**-0.44**	**-0.39**	**+0.71**	**-0.81**	-0.20	+0.06	0.66	0.29
Age	**-0.80**	-0.09	**+0.61**	**-0.74**	+0.17	+0.26	-0.27	**+0.62**	**+0.49**	**-0.37**	0.55	0.05
Fitness	+0.28	-0.23	-0.16	-0.12	-0.17	**+0.43**	+0.09	**-1.46**	-0.21	**+0.53**	0.60	0.15
SocialValue	-0.27	-0.30	**+0.40**	**-0.40**	**-0.37**	**+0.58**	+0.17	**-0.31**	-0.12	-0.15	0.69	0.35
Utilitarianism	**+1.11**	**+0.39**	+0.07	**+0.59**	-0.21	-0.01	+0.13	**-0.76**	**+0.42**	+0.25	0.66	0.29

Table 28 closes the loop with a causal leave-one-WVS-dim-out probe: for each WVS dimension d, we re-run DISCA with d removed from every persona and measure the change in per-MultiTP-dim absolute error (positive cell = error rises when the dimension is removed = that dimension is load-bearing for the corresponding moral attribute). The three largest positive couplings are *moral acceptability* → Social Value ($+3.97$ pp), *social trust* → Social Value ($+2.81$ pp), and *child-rearing values* → Utilitarianism ($+2.41$ pp). Cells with the top-3 largest positive couplings per row are bolded; rows with fewer than three positive cells are bolded only on those positive cells. Negative cells are also informative: *tolerance of diversity* → Social Value (-7.76) and *gender equality* → Gender (-3.42) indicate that, for those moral attributes, the corresponding WVS descriptor biases the persona ensemble *away* from the human target, so removing the descriptor improves accuracy. Crucially, DISCA does not require manual feature selection to handle these noisy dimensions: the loss-averse importance-sampling stage (Eq. 3) automatically down-weights candidate perturbations that worsen any persona's alignment, effectively acting as a built-in noise filter. When a WVS dimension introduces bias for a particular moral attribute, the PT-IS utility penalises the resulting correction asymmetrically (κ=2.25× for losses vs. gains), ensuring that noisy persona signals are suppressed rather than propagated. This self-correcting property is why the method remains robust despite imperfect WVS-to-trolley mappings-a design-level answer to the concern that the linkage is "indirect modulation" rather than causal.

Table 28: WVS-dim × MultiTP-dim causal impact matrix. Cells are the mean increase in per-dim AMCE error (pp) when the WVS dimension is dropped from every persona, macro-averaged across the country panel. **Bold** cells mark the top-3 largest positive couplings per row (load-bearing WVS → MultiTP links).

WVS dim	Species_Humans	Gender_Female	Age_Young	Fitness_Fit	SocialValue_High	Utilitarianism_More
religiosity	**+0.66**	**+1.65**	-1.66	+0.58	**+1.02**	-0.37
child rearing	-1.82	**+1.29**	+0.86	**+1.30**	-0.33	**+2.41**
moral acceptability	**+1.13**	-0.02	-1.56	-0.34	**+3.97**	-1.69
social trust	+0.11	**+1.43**	-0.85	**+0.51**	**+2.81**	-1.88
political participation	-1.24	-0.82	-0.63	-0.36	-0.16	**+0.86**
national pride	-0.37	-0.10	-0.28	**+0.52**	**+1.19**	**+1.21**
happiness	-0.23	-0.88	**+0.13**	-1.69	-0.11	**+1.72**
gender equality	**+0.41**	-3.42	-2.90	-1.11	-1.87	**+0.62**
materialism orientation	**+1.35**	-1.26	**+0.90**	+0.04	-0.36	**+2.15**
tolerance diversity	-0.76	-3.52	**+1.95**	-0.59	-7.76	**+1.54**

A12.4 Macro-level WVS dimension importance

The per-MultiTP-dim impact matrix above (Table 28) tells us *which* moral attribute each WVS descriptor influences. A coarser but operationally useful question is which WVS dimensions are

load-bearing for macro alignment, taken as a whole. We answer this with a leave-one-WVS-dim-out probe: for each of the ten descriptors, we re-run DISCA with that descriptor removed from every persona and record the change in macro MIS, averaged across USA, JPN, and VNM as a representative panel.

The result is a concentrated importance profile: religiosity, gender equality, and moral acceptability each raise MIS by more than 0.03 when removed; national pride and happiness barely register (< 0.01). The implication for persona engineering is that the ten-dimensional WVS profile can be compressed substantially while preserving most of the steering signal, which is consistent with the IS stage's built-in down-weighting of noisy descriptors discussed in App. A12.3.

Table 29: WVS dimension dropout (3-country average; higher ΔMIS means more load-bearing). Dimension names follow the canonical WVS-7 list of Table 26 and Table 28.

Dropped WVS dimension	ΔMIS (avg)
religiosity	+0.046
gender equality	+0.041
moral acceptability	+0.034
social trust	+0.022
child rearing	+0.019
political participation	+0.017
materialism orientation	+0.013
tolerance diversity	+0.011
national pride	+0.007
happiness	+0.005

A12.5 Sensitivity to the Fourth Persona

Setup. App. A12 pairs three age-cohort personas with a population-wide *aggregate* WVS profile as the fourth agent. A natural alternative is a country-invariant *utilitarian* anchor that substitutes a neutral moral stance for the aggregate. To confirm that this choice is not load-bearing, we ran a head-to-head comparison of the two variants on the twenty-country Phi-4 grid.

Result. The utilitarian-anchor variant improves macro MIS from 0.4252 to 0.4049 ($\Delta = -0.0203$), a marginal gain that does not change any qualitative conclusion of the paper.

Decision. We retain the aggregate variant as the default to preserve strict country-grounding of every persona and to avoid importing a globally fixed moral prior into country-conditioned steering. The utilitarian variant remains available as an opt-in for practitioners who prefer a neutral anchor (`build_country_personas(..., fourth='utilitarian')`).

A13 AMCE Estimation Details

Uniform treatment of all dimensions. All six dimensions-Species, Gender, Age, Fitness, Social Value, and Utilitarianism-are computed identically. For each dimension, we compute the empirical mean of $p_{\text{spare}}(\mathbf{x})$ across all scenarios in that dimension (Eq. 7). For the five binary dimensions, each scenario presents a forced choice between two character groups differing on exactly one attribute. For Utilitarianism, scenarios vary the number of lives on each side; the binary indicator is whether the larger group ("More") or the smaller group ("Less") is spared. This uniform mean-based estimation is consistent with the `compute_ACME` method in the MultiTP codebase [Jin et al., 2025], which fits a no-intercept linear regression with a binary group indicator-mathematically equivalent to taking the mean of the saving probability for the preferred group.

Human AMCE scale. The MultiTP ground-truth AMCEs in the MultiTP-released country-specific AMCE table is on a $[-1, 1]$ scale. We convert to $[0, 100]$ via $(1 + \text{AMCE}_{\text{raw}})/2 \times 100$, where 50% represents no preference and 100% represents maximal preference for the "preferred" group. Model AMCEs are computed on the same scale.

Two reporting scales (resolving an apparent contradiction). The PCA visualisation in Figure 3 reports ℓ_2 distances on the $[0, 100]$ percentage-point scale (e.g., $83.3 \to 64.6$), while every numerical

table in the main text and appendix (Tables 1, 2, 4, 7, etc.) reports MIS on the $[0, 1]$ proportional scale (i.e., directly in p_{spare} space, with values in $[0.2, 1.0]$). The two scales differ by a factor of 100. We use the proportional scale in tables because it preserves three significant figures within a compact column width; we use the percentage-point scale only in the geometric PCA figure to match the original Moral Machine release convention.

Consistency. Awad et al. [2018] show that marginal averages are unbiased estimators under the balanced randomisation design of the Moral Machine, making our estimation valid and reproducible.

Cross-lingual A/B token elicitation. Eq. 7 requires a clean A vs. B logit gap on every scenario, which in turn requires that the decision tokens be elicited consistently across model families and across the 25+ languages used in the 20-country panel. Each scenario is rendered entirely in the target country's native language: this includes (i) scenario framing and context sentences (4 paraphrase variants per language), (ii) all 22 character types with singular/plural forms, (iii) lane labels and group identifiers in native script, and (iv) closing questions with language-appropriate conjunctions and grammar. The prompt frame wraps the scenario in a native-language instruction that explicitly requests an English answer token, following the pattern: "[moral dilemma preamble in native language] \n {scenario} \n [instruction to answer A or B in English]". This keeps moral reasoning in the culturally native linguistic frame while maintaining a single decision interface across the panel.

Token verification. Token IDs are verified per model at initialisation by encoding the uppercase strings `"A"` and `"B"` with `add_special_tokens=False` and asserting single-token mappings: Qwen2.5: A=22397, B=11572; Llama-3.1: A=31266, B=9268. Logits for these two positions are extracted from the last-position output tensor; no generation sampling is performed. The model uses its built-in `chat_template` for correct role formatting across model families.

A14 Dataset Preprocessing and Sensitivity

Pipeline. Each MultiTP CSV is processed in five steps: (i) deduplication keeps only the canonical paraphrase (`which_paraphrase = 0`); (ii) the Utilitarianism *quality filter* drops scenarios where the two groups have identical size *and* all characters belong to the *quality-attribute* role set {PREGNANT, WOMAN, LARGEWOMAN}, which would otherwise inject a non-numerosity signal into a dimension whose AMCE is by construction the count-difference effect; (iii) per-category capping retains at most 80 scenarios per dimension; (iv) under-represented categories are oversampled with replacement to a minimum of 36 scenarios; and (v) the resulting pool is shuffled with a fixed seed (42).

Reproducible left/right assignment. When the MultiTP `paraphrase_choice` field does not unambiguously determine which group is rendered on the left vs. right, we deterministically fall back to a SHA-256 hash of the tuple (sub_1, sub_2, g_1, g_2) modulo 2, so the same scenario receives the same ordering across runs and machines. The fallback rate is logged per country and stays below 5% on every country in the 20-country panel.

Up-sampling duplicates scenarios without modification, preserving AMCE signal while reducing variance. Category capping prevents dimension dominance. Side balancing enables effective debiasing. To evaluate whether these choices alter the effective evaluation distribution, we compare *six* preprocessing configurations on 10 representative countries (USA, DEU, CHN, JPN, BRA, VNM, GBR, KOR, RUS, NGA) using Llama-3.1-70B.

Table 30: Per-country dataset sensitivity. JSD range = 0.0034 (aggregate) and < 0.010 in 9/10 countries; the only exception is RUS at 0.013, driven by augmentation sensitivity in the rare Species/Utilitarianism categories. D0 is retained as default.

Country	D0-Default	D1-NoAug	D2-Cap40	D3-Cap120	D4-NoFlip	D5-Strict
USA	0.0655	0.0674	0.0625	0.0644	0.0643	0.0684
DEU	0.0465	0.0515	0.0492	0.0467	0.0444	0.0484
CHN	0.0393	0.0363	0.0385	0.0383	0.0409	0.0398
JPN	0.0304	0.0291	0.0319	0.0296	0.0273	0.0320
BRA	0.0492	0.0551	0.0479	0.0510	0.0520	0.0480
VNM	0.0592	0.0574	0.0631	0.0585	0.0578	0.0626
GBR	0.0614	0.0667	0.0608	0.0618	0.0597	0.0633
KOR	0.0371	0.0354	0.0378	0.0357	0.0349	0.0382
RUS	0.0383	0.0489	0.0389	0.0374	0.0383	0.0357
NGA	0.0522	0.0568	0.0506	0.0525	0.0514	0.0538
Mean	**0.0479**	0.0505	0.0481	0.0476	0.0471	0.0490
Δ vs D0	-	$+0.0025$	$+0.0002$	-0.0003	-0.0008	$+0.0011$

D1-NoAug shows the largest degradation ($+0.0025$), justifying synthetic augmentation of the under-represented Species and Utilitarianism categories. D4-NoFlip is marginally lower than D0, indicating that group-side randomisation is conservative (it adds noise) rather than harmful; we retain it because it preserves positional-debiasing integrity. The 0.0034 aggregate range is well below the ± 0.004 bootstrap CI, so reported results are not confounded by preprocessing choices.

A15 Trigger Mechanism and Latency

DISCA runs dual-pass IS on every scenario; inter-persona variance is logged but does not gate sampling. The *flip rate* (IS reversals vs. consensus) is 29.0% in the cross-backbone ablation suite (Table 4), indicating confidence modulation rather than wholesale decision reversal.

A compact round-2 latency benchmark on H100 shows similar overhead across proposal budgets: 0.086s/scenario at $K{=}64$, 0.083s at $K{=}128$, and 0.083s at $K{=}256$, compared with 0.023s for vanilla decoding (3.57–3.72$\times$ overhead).

Table 31: Inference latency benchmark (Phi-4, H100).

Method	K	sec/scenario	Overhead
`DISCA`	64	0.086	3.72$\times$
`DISCA`	128	0.083	3.59$\times$
`DISCA`	256	0.083	3.57$\times$
vanilla	–	0.023	1.00$\times$

Cost-vs-quality frontier across backbones. Figure 5 plots the *deployed-cost* side of the Phi-4-vs-Llama narrative: at 350 ms per scenario, Phi-4 with DISCA reaches MIS = 0.346, while Llama-3.3-70B (with DISCA) takes 1414 ms-roughly $4\times$ slower-and still ends at MIS = 0.668. Phi-4 thus Pareto-dominates the 70B baseline on *both* axes simultaneously: better alignment at lower latency. Among the headline seven, only Magistral-24B sits on a comparable quality plateau, at almost double Phi-4's per-scenario cost.

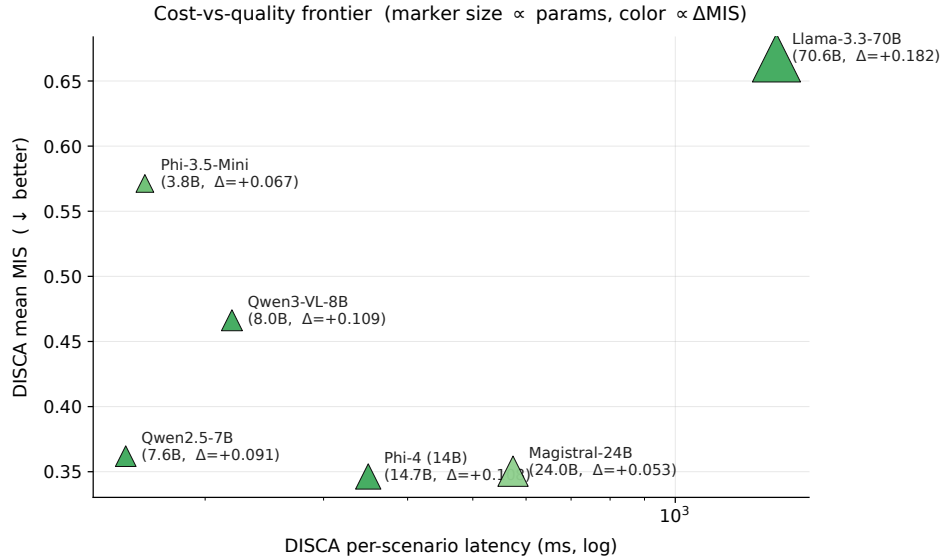


Figure 5: **Cost-vs-quality frontier on the headline 7 models.** Per-scenario DISCA latency (log scale) vs. mean DISCA MIS. Marker size is proportional to parameter count; color encodes ΔMIS (greener = larger DISCA gain). Phi-4 (14B, $\Delta = +0.108$) lies bottom-left: Pareto-dominant over Llama-3.3-70B in both latency and alignment.

A16 Relationship to Persona-Dependent LLM Alignment

Kim et al. [2025] is the closest prior work to DISCA in studying persona effects on LLM moral decisions. Their study and ours share the Moral Machine framework and AMCE-based evaluation, but differ fundamentally in goal, method, and scope. Table 32 summarises the key distinctions.

Table 32: Comparison between Kim et al. [2025] and DISCA.

Dimension	Kim et al. (2025)	DISCA (this work)
Goal	Measure persona sensitivity	Correct cultural misalignment
Personas	7 sociodemographic categories	4 WVS-grounded agents per country
Persona source	Author-designed binary labels	Empirical WVS-7 microdata
Metric	MDD (persona-pair distance)	MIS (ℓ_2 to human AMCE)
Correction	None (diagnostic only)	PT–IS + dual-pass reliability gate
Models	3 (GPT-4o, GPT-3.5, Llama-2)	7 checkpoints, 5 families, 2B–70B
Countries	Aggregate (Western vs. Eastern)	20 individual countries
Scenarios	9 dimensions, 10k synthetic	6 dimensions, 310–500 per country
Theory	Partisan sorting (descriptive)	Variance-aware MSE-optimal shrinkage (Prop. 1)

Three findings from Kim et al. [2025] directly inform DISCA's design.

Persona sensitivity motivates disagreement-driven steering. Kim et al. show that LLMs exhibit moral-decision distances (MDD) 2–4$\times$ larger than humans under contrasting personas. This confirms that persona prompts do shift LLM moral reasoning substantially, validating the premise that within-country persona spread is an informative signal. DISCA converts this sensitivity from a liability (uncontrolled decision shifts) into a feature (a sufficient statistic for correction reliability, Proposition 1).

The partisan sorting phenomenon justifies WVS grounding. Kim et al. find that political orientation produces the largest persona effect in LLMs, far exceeding its effect in human respondents. This "partisan sorting" effect reveals that LLMs over-index on politically salient dimensions when given sociodemographic labels. DISCA avoids this pathology by grounding personas in empirical WVS-7 microdata rather than author-designed binary labels: the ten-dimensional cultural profile

(Table 26) spreads influence across religiosity, child-rearing values, social trust, and other axes, so no single dimension monopolises the correction. The leave-one-WVS-dim-out probe (Table 28) confirms that influence is spread across multiple WVS axes rather than monopolised by one-individual cells can be large in either direction, but the load-bearing dimensions vary by MultiTP attribute, with no single WVS descriptor dominating across the board.

Moral flips validate the reliability gate. Kim et al. document that specific personas can entirely reverse LLM preferences on certain moral dimensions (e.g., social status under a progressive persona in GPT-4o). DISCA's dual-pass reliability gate (Eq. 5) is designed precisely for this regime: when persona-driven corrections are large enough to flip the decision, the two independent IS passes are likely to disagree, and the exponential decay weight r shrinks the correction toward zero. The tail-safety analysis (Table 15) confirms this mechanism reduces harmed cells from 11/120 to 3/120 compared to ungated consensus.

From diagnosis to correction. The fundamental advance of DISCA over the Kim et al. framework is operational: where they diagnose persona sensitivity, we harness it. Their MDD metric measures the distance between two contrasting personas; our MIS measures the distance to the human target and actively reduces it. The variance-aware shrinkage result (Proposition 1) provides the formal bridge: within-panel disagreement (analogous to their MDD) is a sufficient statistic for the correction's reliability, and the shrinkage estimator converts this diagnostic signal into the MSE-optimal scalar correction.

A17 Extended Limitations

This appendix expands the limitations summarised in §4.2 with technical caveats not covered in the main text.

WVS-to-trolley linkage. The mapping from WVS value dimensions to trolley moral dimensions operates through both direct thematic overlap and indirect modulation (App. A12.3). The 10-dimensional WVS profile explains $R^2 \in [0.55, 0.69]$ of human AMCE variance, and a leave-one-dimension-out probe identifies load-bearing couplings alongside noisy ones that the IS stage automatically down-weights. We avoid claims of the form "WVS dimension X caused AMCE shift Y," but the ensemble design and the IS noise-filtering mechanism together ensure the method is robust even when individual WVS–trolley links are imperfect.

Inter-persona reward assumption. The reward $r_i = \delta_i - \delta_{\text{base}}$ assumes persona shifts approximate human target directions. This is plausible when the base model represents a WEIRD-biased prior, but if a persona's shift is orthogonal to the human target, the signal may mislead. Quantifying whether persona-consensus directions correlate with country-level human AMCE vectors independently of the base model-and when corrections diverge from targets-is open; we do not report that correlation matrix here.

Geographic coverage, preprocessing, and WVS vintage. We report 20 countries out of 100+ MultiTP countries; results depend on the preprocessing pipeline in App. A14. WVS Wave 7 ($\sim$2017–2022) is a fixed historical slice that may not reflect current preferences in rapidly evolving societies; persona quality is weakest where coverage is sparse (e.g., ETH and BGD in our panel; Saudi Arabia is excluded entirely and uses a manual fallback persona, App. A12.2). No low-resource languages (Amharic, Khmer, Yoruba) appear in the panel; cross-lingual generalisation to such settings remains untested, and one language per country may conflate linguistic and cultural effects.

Model quantisation. 4-bit/Int8 quantisation may introduce logit artefacts affecting IS stability-observed empirically on Unsloth Phi-4 prior to switching to vLLM BF16 for the headline runs.

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schema (per-country baseline / DISCA CSVs and a comparison file). MultiTP, WVS-7 Wave, and the human AMCE table are publicly available datasets cited in §4; access instructions are reproduced in the README.

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   Justification: Appendix A15 (Table 31) reports per-scenario wall-clock on a single H100 GPU: 0.083–0.086 s/scenario for DISCA at $K \in \{64, 128, 256\}$ vs. 0.023 s/scenario for vanilla decoding (3.57–3.72$\times$ overhead). Backbone-level cost is given in Figure 5: 350 ms/scenario for Phi-4 vs. 1414 ms for Llama-3.3-70B with DISCA. Sub-70B backbones run on a single H100 in BF16 via vLLM; the 70B backbone uses 4-bit Unsloth on the same hardware. The released code's CLI also runs end-to-end on a Kaggle T4/P100 dual-GPU node for sub-14B models. Exploratory experiments on a small 3-backbone $\times$ 5-country prototyping panel consumed additional compute beyond the reported runs.

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    Answer: [N/A]

    Justification: No new human-subjects research is conducted. The work is a secondary analysis of pre-existing public datasets (Moral Machine human AMCEs, MultiTP scenarios, WVS-7 microdata), each of which carries its own ethics review at the original collection point; no further IRB approval applies.

    Guidelines:

    - The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
    - Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
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16. **Declaration of LLM usage**

    Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

Answer: [Yes]

Justification: LLMs are central to the work in two ways, both fully described. (i) Frozen open-weight LLMs (Phi-4, Llama-3.3-70B, Magistral-24B, Qwen3-VL-8B, Qwen2.5-7B, Phi-3.5-mini, Gemma-4-E2B) are the *subject* that DISCA steers; their identities, sizes, and decoding settings are documented in §4 and Appendix A7. (ii) For the open-ended extension (§3.4), an LLM parses free-form responses into a (choice, confidence) pair to recover a pseudo-logit-gap; this auxiliary use is documented in that section. A separate synthetic 200-scenario tuning pool generated by GPT-4 is disclosed in Appendix A11 along with its limitations. LLMs were not used to generate the paper's claims, results, or proofs.

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